
Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

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Abstract

Generative image models trained on large-scale datasets can achieve broad visual coverage, yet they still fail on concepts that plausibly occur in their training distributions, including human hands and objects appearing in groups of four. Because such failures have often been reported as isolated examples, it remains unclear whether they are unusual accidents of particular prompts or evidence of a more general mismatch between data distributions and model generations. We address this question by introducing a systematic approach for identifying and characterizing “conceptual blindspots”: concepts that are present in the data distribution but absent, suppressed, or misrepresented in generated images. Our method uses sparse autoencoders (SAEs) to obtain interpretable concept embeddings, making it possible to compare concept prevalence quantitatively across real and generated image sets. We train an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts—the largest such SAE to date—so that discrepancies can be studied at a fine conceptual resolution. Across four widely used generative models (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky), the same framework identifies both suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated blindspots (e.g., wood background texture and palm trees). At the individual datapoint level, the analysis also isolates memorization artifacts, where models reproduce highly specific visual templates encountered during training. Overall, we propose a theoretically grounded framework for assessing conceptual fidelity with respect to an underlying data-generating process, while demonstrating its use on contemporary text-to-image systems.

1 Introduction

Generative image models trained on large-scale datasets now support a wide range of visual applications [1–9] and are increasingly used in settings beyond conventional image synthesis [9, 15]. Even with this broad capability, qualitative [16, 18] and quantitative studies [19, 20] show that models can struggle with concepts that should be well represented in natural image corpora, such as human hands [21–24], objects appearing in groups of four [25], and negations or object relations [20, 26]. When prompted for these concepts, models often generate semantically nearby outputs without realizing the intended concept precisely, for example rendering a hand with six fingers. We refer to these discrepancies as “conceptual blindspots”¹ cases where exposure to relevant visual evidence does not translate into reliable generation of the corresponding concept. The central question is therefore broader than any single failure case: do these examples reflect isolated quirks of specific prompts, or do they reveal a systematic tendency for generative models to suppress or overproduce concepts across a wider conceptual space?

¹We borrow the term “blindspots” from psychology literature [27], wherein it is used to refer to scenarios where an agent makes biased decisions despite exposure to observations that contradicts their decision’s rationale.



Figure 1: $\delta(k)$ quantifies a model’s tendency to over- or under-generate a concept c_k compared to its natural-data frequency. We deem concepts with $\delta(k) < 0.1$ as **suppressed conceptual blindspots** and concepts with $\delta(k) > 0.9$ as **exaggerated conceptual blindspots**. The depicted images, generated by four popular generative image models, show examples of images with conceptual blindspots, as well as aligned concepts.

Answering this question is difficult with standard evaluation tools for generative image models [28, 29]. Many established metrics emphasize aggregate image realism, such as FID [30], and therefore can miss concept-specific distributional failures. Other approaches, including CLIPScore, diversity measures, and distribution-coverage statistics [31–33], provide useful model-level signals but do not localize mismatches to fine-grained visual features or concepts [34, 35]. Qualitative evaluations such as participant surveys [5, 17, 36–38] and open-ended exploration [39] can reveal individual failures, but they are harder to repeat systematically across datasets, benchmarks, and model families.

This work. Motivated by these gaps, we argue that studying conceptual blindspots requires an automated and unsupervised methodology that identifies concepts whose generation odds differ between the true data-generating process and the trained model. Our contributions in this work are three-fold:

- **Formalizing Conceptual Blindspots in Generative Image Models.** We introduce a framework for identifying and quantifying conceptual blindspots by comparing generated images to natural-image concept distributions (Section 3). This formalization generalizes beyond anecdotal examples or hand-selected concepts while remaining grounded in measurable distributional mismatch.
- **A Scalable, Unsupervised Approach for Identifying Conceptual Blindspots using Sparse Autoencoders.** We develop a methodology that uses sparse autoencoders (SAEs) to extract and compare concept distributions between natural and synthesized images (Section 4). For this purpose, we train and open-source an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts, the largest such RA-SAE to date.
- **Exploratory Tool and Analysis.** Our exploratory web tool supports distribution- and datapoint-level analyses of blindspots across models (Sections 5.1–5.4). We apply the method to Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1 (Section 5), identifying suppressed conceptual blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated conceptual blindspots (e.g., wood background texture and palm trees), as shown in Fig. 1.

The model, code, and web tool are available at a software repository.

2 Related Work

2.1 Explainability in Vision

Early explainability work in computer vision focused on attributing predictions to influential input regions [40–42]. These methods are broadly useful for localization, but they often provide limited semantic information about learned representations and can yield unreliable explanations [43–45]. Concept-based interpretability [46] addresses this broader representational question by identifying semantically meaningful directions in neural networks, revealing not only where models attend but also what concepts and structures they encode [47–49].

Recent work shows that several concept-based interpretability methods—ACE [44], CRAFT [48], and Sparse Autoencoders (SAEs) [50, 51]—can be understood as variants of a shared dictionary-learning problem with different constraints [52]. Within this family, SAEs are especially attractive for large-scale concept discovery because they provide sparse, reusable concept bases. Although

76 standard SAEs can produce overly specific features [53], struggle with compositionality [54], or have
77 limited intervention effects [55], newer variants such as archetypal SAE (RA-SAE) [56] improve
78 their suitability for broad concept-level analysis.

79 Beyond SAEs, prompt-based probing [57], attention-map or activation visualizations [58, 59], and
80 dataset-level statistics [32, 33] also provide useful views into model behavior, but each offers
81 only partial evidence for concept-level blindspots. Their main limitation is granularity: they often
82 operate over whole images, prompts, or global distributions rather than the fine-grained features and
83 concepts where suppression or exaggeration may occur [34, 35]. They can also depend on subjective
84 interpretation and may not distinguish among different failure modes [60]. Existing methods and
85 metrics therefore leave room for a systematic way to identify feature- and concept-level weaknesses
86 across generative image models [28].

87 2.2 Generative Image Models

88 Diffusion-based methods now span several generative vision modalities, including image [4, 61–64],
89 video [65–72], and 3D [7, 73–75]. For image generation, this trajectory begins with denoising
90 diffusion probabilistic models (DDPMs) [76] and extends to non-Markov diffusion processes such as
91 denoising diffusion implicit models (DDIMs) [77].

92 The Stable Diffusion (SD) [9] family made DDPM-style image generation broadly accessible to both
93 research and open-source communities. Subsequent versions, including SD 2 [78], SD 3 [79], and SD
94 XL [80], expanded this line, while architectural extensions introduced constraints for style [81, 82],
95 pose [83], identity [84, 85], and additional input modalities such as image-to-image generation.
96 Related latent diffusion systems, including Kandinsky [86], PixArt [87], DeepFloyd [4, 88], and
97 FLUX [89], further broaden the model classes for which concept-level diagnostics are relevant.

98 2.3 Datasets for Generative Image Models

99 The recent success of generative vision models is closely tied to increased compute and large-scale
100 internet datasets [90–92]. LAION-5B [93], scraped from Common Crawl [94], has been especially
101 important for open-source text-to-image systems, including SD and its derivatives; it contains more
102 than 5B image-caption pairs, 2.3B of which are in English. Other widely used resources, such
103 as COYO-700M [95] and Conceptual Captions [96], provide 700M and 3M image-caption pairs,
104 respectively, illustrating the broader data regime in which concept coverage becomes a central
105 concern.

106 As LAION-5B became widely used, researchers documented biases and safety issues in the
107 dataset [97–101]. Filtering did not remove all harmful content [97–99], including NSFW material
108 [100] and hundreds of CSAM instances [101], which prompted its temporary removal from
109 official channels. In addition, low-quality images [102] and internet-style captions such as product
110 descriptions can differ from how users prompt trained models [103], making dataset-level concept
111 distributions an important object of analysis.

112 2.4 Concept Discovery and Sparse Coding in Generative Image Models

113 Dictionary learning seeks sparse representations in which each sample is reconstructed from a
114 small number of dictionary atoms [104, 106]. Building on compressed sensing theory [107, 108],
115 the field has developed from vector quantization [109] to methods such as Non-negative Matrix
116 Factorization [110, 111], Sparse PCA [112], and K-SVD [113]. More recent work includes online
117 methods [114], structured sparsity [115], theoretical guarantees [116, 117], and connections to deep
118 learning [118, 119], making sparse coding a natural foundation for broad concept discovery.

119 Sparse coding has also been used to study the emergence of high-level concepts inside diffusion
120 models [120]. Before diffusion models, concept-grounded interpretability was applied to generative
121 architectures through concept-bottleneck models, which require human intervention during training
122 [121], and post-hoc detectors that retrofit concept supervision [122]. Because both approaches
123 depend on human-defined concepts, they can miss broader trends that are not specified in advance.

124 Aggregate metrics, including precision, recall, density, and coverage [35, 123], as well as latent density
125 scores that predict sample quality from a model’s latent space [124], provide scalable evaluations

of generative image models. These methods can uncover distributional gaps, but they offer limited insight into which specific concepts are under- or over-represented.

Together, these limitations motivate a scalable, unsupervised framework for identifying and quantifying concept-level failure modes in generative image models [125], while preserving enough resolution to support model comparison, dataset analysis, and targeted inspection.

3 Formalizing Conceptual Blindspots in Generative Models

We formalize *conceptual blindspots* as systematic discrepancies between the conceptual content of natural images and the conceptual content of model-generated outputs. This framing yields quantitative measures for identifying concepts that are under- or over-represented by a generative model relative to the data distribution it is intended to match. The process is illustrated in Fig. 2. Our derivation uses standard assumptions from representation learning and disentanglement [126, 130], but the empirical analyses below show meaningful phenomena even when these assumptions hold only approximately.

Definition 1 (Data-Generating Process). Let $\mathcal{C} \subset \mathbb{R}^K$ denote a latent space with a Boltzmann prior $p(c) = \exp(-E(c))Z^{-1}$, where $E(\cdot)$ denotes an energy function that linearly decomposes over individual latents and Z is the corresponding partition function, i.e., $E(c) = \sum_k E(c_k)$ and hence $p(c) = \prod_k p_k(c_k)$. A data-generating process (DGP) is an invertible function $G: \mathcal{C} \rightarrow \mathcal{X}$ that maps the latents $c \in \mathcal{C}$ to observations $x \in \mathcal{X}$, i.e., $x = G(c)$.

For notational simplicity, $p(\cdot)$ denotes both the latent density $p(c)$ and its pushforward to image space $p(x)$, where $x = G(c)$. This shared notation is valid because the invertibility of G induces a distribution over $\mathcal{X}$ through the change of variables formula. In this formulation, individual latent dimensions represent the **Concepts** underlying the image distribution $P_{\mathcal{X}}$ over the observation space $\mathcal{X}$. These concepts may correspond to properties such as color, shape, size, or location, among many other visual factors [131, 132]. We also allow the data-generating process to associate a text description $t \in \mathcal{T}$ with each sampled image, while leaving the caption-generation mechanism implicit. A text-conditioned **Generative image model** g_{θ} , with parameters θ , can then be trained on image-text pairs to map a noise signal $\eta \sim \mathcal{N}(0, I)$ and scene description t to an image x intended to depict that description.

To define conceptual blindspots for g_{θ} , we compare the probability mass assigned to a concept by the data-generating process with the mass reflected in the model’s generations. Operationally, we begin from an evaluation set of natural image-text pairs $(\mathcal{D}_{\mathcal{X}}, \mathcal{D}_{\mathcal{T}})$ and generate a corresponding synthetic image set $\mathcal{D}'_{\mathcal{X}}$ from the same text descriptions. To estimate concept occurrence, we use an **Energy model** $\xi: \mathcal{X} \rightarrow \mathbb{R}^d$ that maps each image to a d -dimensional vector, with each dimension representing the energy of the k^{th} concept, i.e., $\xi_k(x) = E(c_k)$. The invertibility assumption on the data-generating process makes these estimates meaningful in the formal model. The probability assigned to image x can then be written as $p_k(x) \propto \exp(-\sum_k \xi_k(x))$ (where Z_{ξ} is the partition function), yielding a dataset-level estimate $p_k(\mathcal{D}_{\mathcal{X}}) = \prod_{x \in \mathcal{D}_{\mathcal{X}}} p_k(x)$. Using this quantity and the sigmoid map $\sigma(\cdot)$, we define the average energy difference for the k^{th} concept between $\mathcal{D}_{\mathcal{X}}$ and $\mathcal{D}'_{\mathcal{X}}$.

Definition 2 (Energy Difference). Let $x \in \mathcal{D}_{\mathcal{X}}$ denote a real image sampled from the data-generating process G , and let $x' \in \mathcal{D}'_{\mathcal{X}}$ be a synthetic image generated by the model g_{θ} . Let $\xi_k: \mathcal{X} \rightarrow \mathbb{R}$ denote the energy assigned to the k^{th} concept by the energy model ξ . We define the

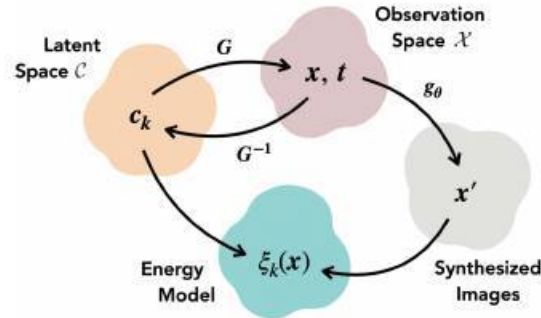


Figure 2: **Intuitive mapping of our framework.** Latent concepts $c_k \in \mathcal{C}$ are mapped to observations (x, t) through the (invertible) DGP, while S_{θ} generates images x' from t . The energy model $\xi_k(x)$ extracts concept representations from both x and x' , allowing concept distributions to be compared across natural and generated data.

178 energy difference for concept k as:

$$\begin{aligned}\delta_{g_\theta \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{\mathbf{x}'}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x}}[\xi_k(\mathbf{x})]) \\ &= \frac{p_k(\mathcal{D}'_{\mathcal{X}})}{p_k(\mathcal{D}_{\mathcal{X}}) + p_k(\mathcal{D}'_{\mathcal{X}})},\end{aligned}\quad (1)$$

179 where the expectations are taken over $\mathcal{D}'_{\mathcal{X}}$ and $\mathcal{D}_{\mathcal{X}}$, respectively, and $p_k(\mathcal{D}) \propto \exp(-\sum_{\mathbf{x} \in \mathcal{D}} \xi_k(\mathbf{x}))$
180 denotes the unnormalized conceptual probability mass of dataset $\mathcal{D}$ under concept k .

181 Thus, the energy difference for concept k summarizes how strongly that concept appears in one set
182 of observations (here, $\mathcal{D}'_{\mathcal{X}}$) relative to a baseline dataset (here, $\mathcal{D}_{\mathcal{X}}$). This comparison supports the
183 following definition of conceptual blindspots.

184 **Definition 3 (Suppressed / Exaggerated Conceptual Blindspots).** Given a generative image model
185 g_θ , we say, compared to the data-generating process G , c_k is a suppressed conceptual blindspot in
186 the model if $\delta_{g_\theta \leftrightarrow G}(k) < \lambda_{\min}$ and exaggerated if $\delta_{g_\theta \leftrightarrow G}(k) > \lambda_{\max}$.

187 Overall, a conceptual blindspot is any concept whose occurrence in generated images deviates sub-
188 stantially from its prevalence under the data-generating process, either by suppression or exaggeration.
189 Suppressed concepts have disproportionately low activation (e.g., $\delta(k) < \lambda_{\min}$), whereas exaggerated
190 concepts are overrepresented (e.g., $\delta(k) > \lambda_{\max}$). In the experiments, we set $\lambda_{\min} = 0.1$ and
191 $\lambda_{\max} = 0.9$ to isolate these two regimes.

192 This definition is related to “mode collapse” in prior work (e.g., Bau et al. [39]), but it shifts the unit
193 of analysis. Mode collapse concerns exaggerated or suppressed odds of generating *entire images*,
194 whereas our definition measures changed odds for specific concepts within images. For example, if a
195 model fails to produce images of an object with a white background, we classify that concept as a
196 suppressed conceptual blindspot.

197 4 Method: Operationalizing the Definition of Conceptual Blindspots

198 We next describe the pipeline used to identify conceptual blindspots in a generative model g_θ .
199 Following Sec. 3 the required ingredients are (i) generated images from g_θ that can be compared
200 with the reference data-generating process, and (ii) an energy model that places both real and
201 generated images in a shared concept space. Below, we use $\|\cdot\|_F$ to denote the Frobenius norm
202 and $\|\cdot\|_0$ to denote the number of non-zero entries (the ℓ_0 pseudo-norm). For a vector or matrix $\mathbf{X}$,
203 $\mathbf{X} \geq 0$ implies element-wise non-negativity. For $n > 0$, we let $[n] := \{1, \dots, n\}$, and denote the i -th
204 row of a matrix $\mathbf{A}$ by $\mathbf{A}_i$.

205 **From Prompts to Latent Representations.** To identify conceptual blindspots in g_θ , we compare
206 natural image-caption pairs $(\mathbf{x}, t)$ from $\mathcal{D}_{\mathcal{X}}$ with synthetic counterparts generated by g_θ from the
207 same text descriptions.

208 Given t , we synthesize a counterpart image $\mathbf{x}'$ with a pretrained text-to-image generator $g_\theta : \mathcal{T} \rightarrow \mathcal{X}$,
209 implemented as a denoising diffusion probabilistic model (DDPM) [78, 86, 87]. Sampling occurs in
210 latent space via a reverse trajectory $(\gamma_t)_{t=0}^T$:

$$\gamma_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\gamma_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\gamma_t, t, c) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

211 where $\alpha_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ follow the standard cosine noise schedule. The final latent γ_0
212 is decoded with a pretrained VAE to obtain $\mathbf{x}' = \text{VAE}(\gamma_0)$. For the rest of the paper, we treat g_θ as
213 a black-box prompt-to-image map, $t \mapsto \mathbf{x}'$, so the diagnostic can be applied across the evaluated
214 model classes without relying on internal access.

215 **Defining the Energy Model.** Prior work suggests that self-supervised representations can invert
216 aspects of the data-generating process and identify the underlying energy function up to linear
217 transformations [126, 128, 133, 134]. We therefore use DINOv2 [135], a state-of-the-art self-
218 supervised vision model, as the feature backbone for our analysis. Because the number of concepts
219 in the DGP may exceed the dimensionality of the backbone feature space [51, 136], we train sparse
220 autoencoders (SAEs) on DINOv2 features to identify concept-aligned subspaces [50, 56, 137–139].
221 The intuition is that if DINOv2 represents generative concepts with approximately orthogonal

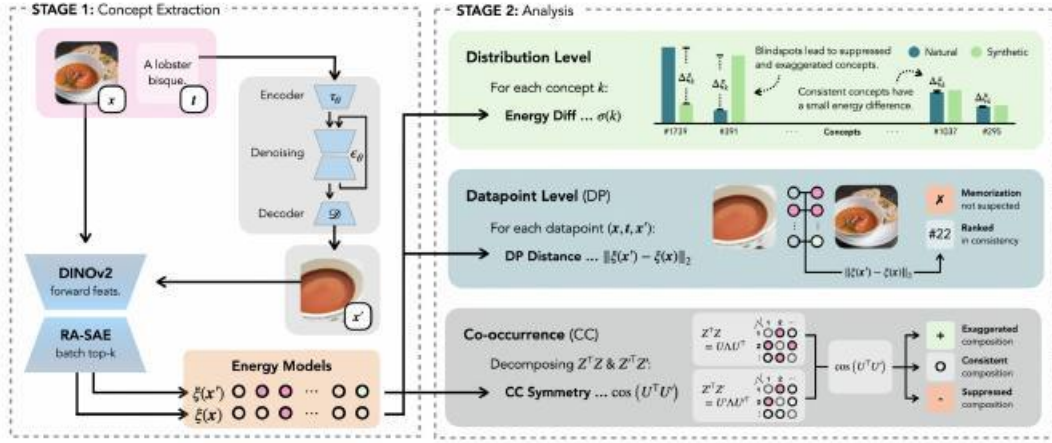


Figure 3: **Concept Extraction Pipeline.** For a triplet (x, t, x') , we extract DINOv2 features from the natural and generated images and process them with RA-SAE to obtain sparse concept embeddings. These embeddings yield energy models $\xi(x)$ and $\xi(x')$ in a shared concept basis; in particular, $\xi_k(x) = E(c_k)$ holds the energy in the k^{th} concept.

directions, as reflected in the independence constraint in Def. 1, then an SAE can separate those concepts into individual latent dimensions [136]. The activation of each SAE dimension then serves as our approximation to the ground-truth energy assigned to the corresponding concept.

Formally, let $f : \mathcal{X} \rightarrow \mathbb{R}^d$ denote the DINOv2 feature extractor, and let $a = f(x) \in \mathbb{R}^d$ be the feature vector for either a natural or synthetic image from $\mathcal{D}_{\mathcal{X}}$ or $\mathcal{D}'_{\mathcal{X}}$. For n image-text pairs, we stack the real and generated features into matrices $A, A' \in \mathbb{R}^{n \times d}$. The SAE then decomposes each feature vector into a sparse combination of learned concept atoms. Specifically, let $D \in \mathbb{R}^{d \times K'}$ denote a dictionary of K' concept vectors, and let $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^{K'}$ be the SAE encoder that maps input features to sparse codes. Applying Ψ row-wise yields the matrix of activations $Z = \Psi(A) \in \mathbb{R}^{n \times K'}$, where each row $z_i = \Psi(a_i)$ represents the concept decomposition of an image. The decoder reconstructs features via ZD^T , and the SAE is trained to minimize the reconstruction error subject to sparsity and non-negativity:

$$\min_{\Psi, D} \|A - \Psi(A)D^T\|_F^2 \quad \text{s.t.} \quad \Psi(A) \geq 0, \|\Psi(A)_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

Vanilla SAEs can learn arbitrarily oriented dictionaries, which makes downstream concept analyses sensitive to random initialization. To reduce this instability and make the diagnostic more reproducible across experimental runs, we use the *Archetypal* SAE (RA-SAE) [56] with a TOP-K sparsity constraint [137]. RA-SAE constrains the dictionary D to be a convex combination of training data. Specifically, we write $D = WA$ with $W \in \Omega_{K', n}$, the set of row-stochastic matrices in $\mathbb{R}^{K' \times n}$:

$$\Omega_{K', n} := \left\{ W \in \mathbb{R}^{K' \times n} \mid W \geq 0, W\mathbf{1} = \mathbf{1} \right\}. \quad (3)$$

Thus every atom D_i lies in the convex hull of the data $\text{conv}(A)$, and any reconstruction ZD^T resides inside the conic hull of the data $\text{cone}(A)$. This constraint keeps learned concepts anchored to the support of the data distribution [56]. After training, the SAE yields a consistent set of sparse codes: Z for real images and Z' for their generated counterparts. *These codes describe prompt-conditioned visual semantics through a shared set of interpretable concepts, with concept activations serving as energy values for the blindspot analysis.* In summary, given a prompt and associated real image (t, x) , the method produces sparse concept vectors (z, z') that make real and generated visual content directly comparable in a common conceptual basis.

The full pipeline therefore starts from a captioned image (t, x) , synthesizes a counterpart x' , and maps both images into a shared sparse concept space using a vision encoder and trained SAE. The resulting interpretable representations (z, z') support concept-level evaluation of shifts induced by the generative process.

5 Results

We evaluate four generative image models trained on LAION-5B—SD 1.5, SD 2.1, PixArt, and Kandinsky—using $|\mathcal{X}| = 10,000$ image-text pairs and their corresponding generations (Fig. 4). The analysis spans three levels (Fig. 3): a **distribution-level** evaluation identifies suppressed and exaggerated concepts; a **datapoint-level** analysis surfaces failures tied to ambiguity, omission, and memorization; and a **compositional** analysis probes distortions in concept co-occurrence geometry.

A core contribution is the interactive exploratory tool shown in Fig. 5. The web interface is built around a UMAP projection of concept representations, enabling visualization and comparison of concept-level energy differences across evaluated models. It is publicly available at a software repository, together with pre-computed energy difference data for SD 1.5, SD 2.1, PixArt, and Kandinsky. The analyses below use this tool to move between broad distributional summaries and targeted concept inspection. Its primary functionalities include:

- **Contrast different models and architectures.** For each evaluated model, the tool provides a UMAP visualization over all 32,000 RA-SAE concepts, with each point representing one concept and color indicating its energy difference.
- **Inspect concepts.** Each concept card reports key statistics, representative real and generated images (x , x'), and visualized co-occurrence patterns.
- **Explore blindspots.** Beyond UMAP and per-concept views, global rankings of suppressed and exaggerated blindspots help identify concepts that merit closer inspection.

Rather than exhaustively studying a single phenomenon, we present findings that illustrate how the same tool can support broad, customizable exploration of concept-level behavior.



Figure 4: **Representative image-caption pairs** from LAION-5B and matching generations produced from the same prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

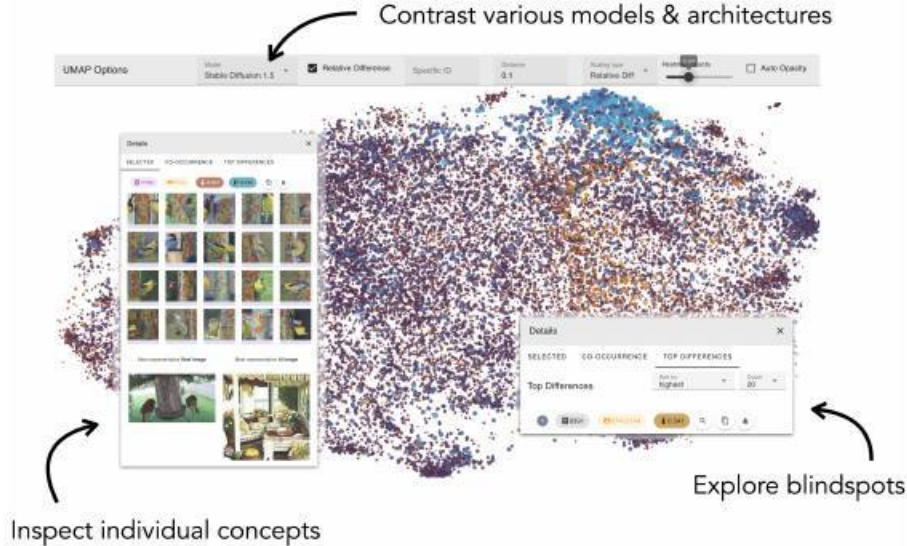


Figure 5: **Overview of the Exploratory Tool.** The web interface displays a UMAP projection for each evaluated model, where each dot is a concept color-coded by its energy difference. Selecting a concept opens a detail panel with illustrative images, statistics, and representative natural and generated images x and x' . Co-occurrence rankings are shown alongside global rankings of suppressed and exaggerated blindspots.

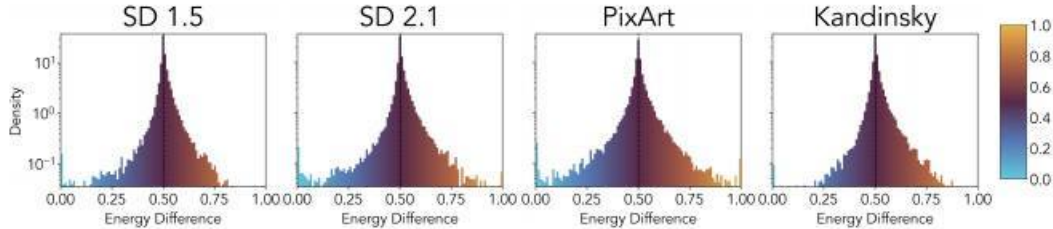


Figure 6: **Concept Energy Distribution.** Log-scale histograms of energy differences $\delta(k)$ across 32,000 concepts, comparing natural and synthesized distributions for each evaluated model. Values left of zero indicate suppressed concepts, while values right of zero indicate exaggerated concepts.

5.1 Generative Image Models Cannot Produce Unseen Concepts

We first assess disparities between generated and natural image distributions by measuring the marginal energy difference $\delta(k)$ over the 32,000 RA-SAE concepts. As defined in Sec. 4, this score captures the relative prevalence of each concept in synthesized versus natural image sets. A value of $\delta(k) < 0.1$ indicates that concept k is under-represented in generated images, whereas $\delta(k) > 0.9$ indicates over-representation. Fig. 6 shows the distribution of $\delta(k)$ for each of the four evaluated models. All models exhibit heavy-tailed histograms with substantial mass near both extremes, indicating systematic concept-coverage discrepancies rather than isolated failures. The left tail, corresponding to suppressed concepts, is denser and longer than the right tail, indicating a consistent tendency toward concept suppression. This asymmetry appears in the negative skewness values: Skewness = -0.54 for SD 2.1, -0.40 for both SD 1.5 and PixArt, and -0.23 for Kandinsky.

Although every evaluated model contains both suppressed and exaggerated concepts, the detailed profiles differ across architectures. For instance, PixArt shows a wider spread, suggesting stronger variation in concept representation. At the same time, the shared left-skew across all distributions points to a broader tendency toward concept omission, whose structure and specificity we examine next.

5.2 Structure and Specificity of Conceptual Blindspots

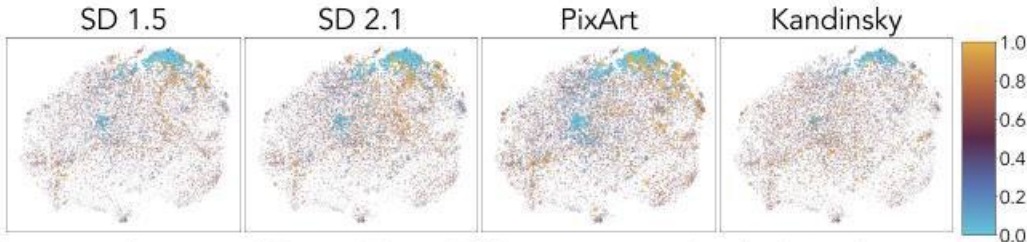


Figure 7: **Structure of Concept Energy Differences.** UMAP visualizations of 32,000 concepts, colored by their energy difference $\delta(k)$ between natural and synthesized distributions. Clusters reveal structured blindspot patterns, with suppressed concepts shown in blue and exaggerated concepts shown in yellow.

Having quantified marginal concept-frequency discrepancies, we next ask whether these discrepancies have global structure. We embed all 32,000 concepts into two dimensions with UMAP on the sparse codes and color each concept by its $\delta(\cdot)$ value. As Fig. 7 shows, distinct concept clusters appear for all evaluated models. These clusters often align with contiguous blindspot regions, especially among suppressed (blue) concepts, suggesting that blindspots are structured and may reflect shared influences from training distributions or architectural priors. To measure how consistently these patterns transfer across models, we compare both the magnitude and structure of concept-level $\delta(\cdot)$ values. Fig. 8 reports scatter plots and pairwise Pearson correlations between the $\delta(k)$ vectors of SD 1.5 and the other models. SD 1.5 and 2.1 exhibit strong correlation ($r = 0.82$), consistent with their related architectures and training pipelines. Their correlations with PixArt and Kandinsky are lower— $r = 0.41$ and $r = 0.46$, respectively—indicating that different model families emphasize different regions of the concept space.

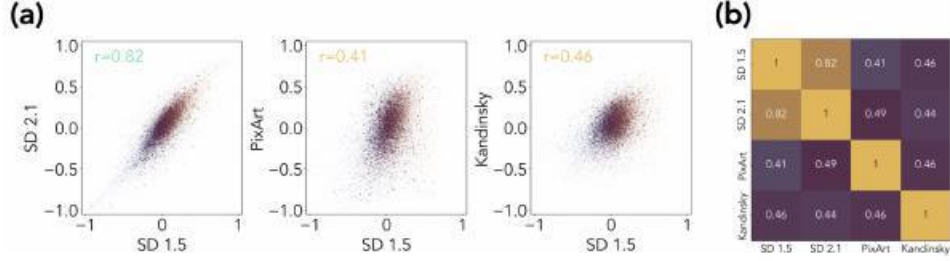


Figure 8: **● Cross-Model Concept Energy Correlation.** Pairwise scatter plots of $\delta(k)$ across the four evaluated models, with Pearson correlation coefficients reported at top left. Strong alignment between SD 1.5 and SD 2.1 contrasts with weaker correlations for other architectures, indicating model-specific blindspot profiles. **● Correlation Matrix of Conceptual Blindspots.** Heatmap of pairwise Pearson correlations for δ across models, quantifying how much conceptual blindspot structure is shared.

Overall, these analyses show that some blindspots are shared across models, plausibly reflecting dataset-level factors, while others are model-specific and may arise from training dynamics, architecture, or capacity. This motivates studying both cross-model blindspots and model-unique blindspots in the qualitative analyses that follow.

5.3 Qualitative Examples of Blindspots

We next inspect concrete examples of suppressed and exaggerated blindspots to clarify what kinds of concepts fall into these regimes. Fig. 9a shows a suppressed blindspot shared by all evaluated models: none reproduce the concept solid white on documents. Even when the caption references the relevant visual content, the generated images do not capture the intended semantics, suggesting that this concept-space region is under-sampled across models. Fig. 9b shows a model-specific blindspot: the concept pan is captured by three models but is missing from Kandinsky generations. This example mirrors Sec. 5.2 where some blindspot structure is shared while other regions vary by model.

Conversely, Fig. 10 shows an exaggerated blindspot in which shadow under animal is over-emphasized in generated images. The shadows are plausible in isolation, but their consistent and pronounced rendering across models, compared with the more variable appearances in natural images, suggests an overactive prior. We did not find clear examples of concepts uniquely exaggerated by only one evaluated model, which suggests that the strongest exaggerations in this analysis are more shared than model-specific.

Together, these examples show how conceptual blindspots appear in concrete generations and demonstrate that the energy-based diagnostic can surface both shared and model-specific failures. The consistent difficulty with clear or solid background elements, such as whitespaces

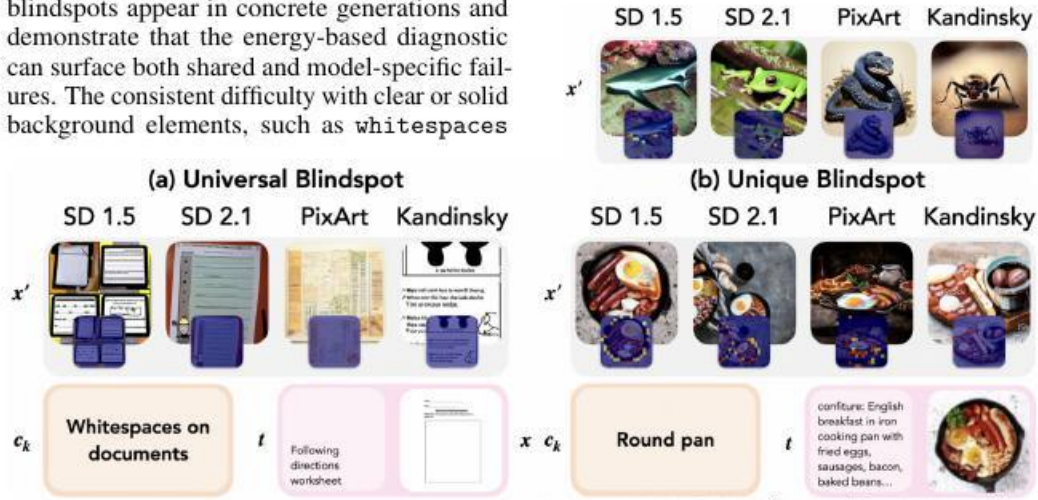


Figure 9: **● Examples of Suppressed Conceptual Blindspots.** Natural images x , representative captions t are shown alongside four synthesized images x' generated using S_θ . The universal blindspot appears in all evaluated models, while the model-specific blindspot appears only in Kandinsky.

on documents, indicates that some limitations may reflect broader architectural constraints or training-data biases rather than only idiosyncratic model behavior.

While these aggregate analyses are informative, they also raise a finer-grained question: do blindspots appear only after aggregating many samples, or are they visible at the level of individual datapoints? Datapoint-level inspection helps distinguish among prompt misinterpretation, omission, memorization, and other mechanisms behind the measured discrepancies.

5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

To move beyond population-level statistics, we examine individual natural and generated image pairs whose concept-averaged $\delta(\cdot)$ values have the largest and smallest differences. This analysis makes model successes and failures interpretable at the sample level, with failures often arising from prompt ambiguity or memorization artifacts. Fig. 11a shows cases with near-zero differences in average $\delta(\cdot)$ values. In these cases, the generated images are conceptually close to the originals. Qualitative inspection, however, suggests that this closeness can result from replication of memorized templates rather than faithful prompt understanding: repetitive visual structures, such as clothing outlines or object arrangements, indicate copying from frequent training patterns. By contrast, Fig. 11b shows samples with some of the largest $\delta(\cdot)$ values. These examples reveal substantial conceptual divergence between synthesized and natural images. Some discrepancies arise from underspecified or noisy captions, but others reveal genuine blindspots: the prompt describes a clear concept present in $\mathcal{D}_{\mathcal{X}}$, yet the model does not realize it in $\mathcal{D}'_{\mathcal{X}}$. Such cases suggest that even when the language signal is sufficient, some concepts remain difficult for the model to generate.



Figure 11: **• Datapoint-level Conceptual Alignment.** (a) Examples with minimal energy differences, where generations appear to reuse training-like patterns. (b) Examples with large differences, where prompt ambiguity or model limitations produce substantial concept divergence.

376 5.5 Analyzing Post-Training Effects

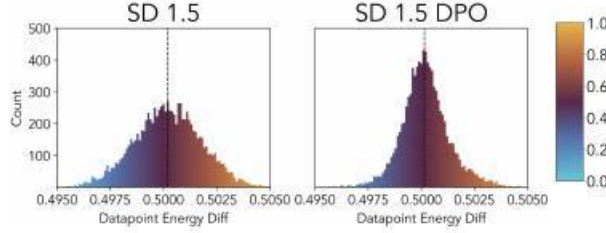


Figure 12: **Effect of DPO on Concept Fidelity.** Histogram of datapoint-wise energy differences between synthesized and natural distributions for SD 1.5 variants with and without DPO.

377 Post-training protocols, including safety fine-tuning, have been argued to reduce the diversity of
 378 model generations [140]. Because our pipeline measures differences between model generations and
 379 the reference DGP in concept space, it can also be used to study post-training interventions such
 380 as DPO, a widely used safety fine-tuning protocol [141]. We compare two SD 1.5 variants: one
 381 trained with DPO and one without DPO. For each image pair $(\mathcal{D}_X, \mathcal{D}'_X)$, we compute the ℓ_2 norm
 382 between their internal concept energy vectors, $\|\xi(\mathcal{D}'_X) - \xi(\mathcal{D}_X)\|_2$. Fig. 12 shows the histogram of
 383 these datapoint-wise energy differences. The DPO-enhanced model has a lower median and narrower
 384 spread, indicating that generated concepts are more consistently aligned with the ground-truth DGP
 385 under this evaluation. This suggests that DPO may regularize the model’s concept distribution,
 386 encouraging outputs that better match the semantic content of the paired natural inputs. Although
 387 the analysis does not isolate the specific inductive biases introduced by DPO, the results indicate
 388 that an objective favoring human-preferred generations can indirectly improve alignment with the
 389 training distribution. In particular, it reduces *both over and under* activation of individual concepts
 390 relative to the natural baseline. These findings show that the pipeline can characterize downstream
 391 effects of post-training interventions not only through output quality, but also through changes in the
 392 conceptual geometry of the model’s output space.

393 5.6 Conceptual Misalignment as a Function of Empirical Frequency

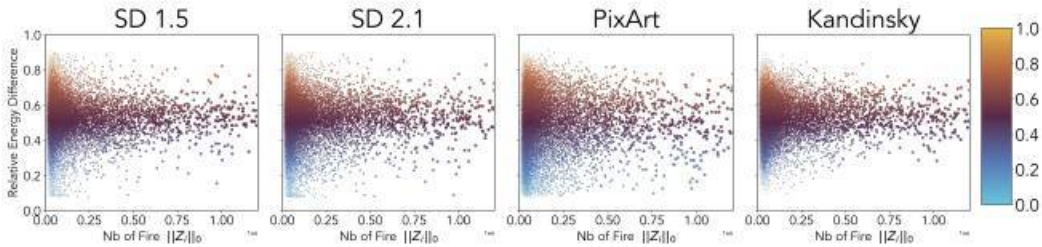


Figure 13: **Concept Fidelity Across Frequency Spectrum.** Scatter plots show the relationship between concept frequency (x-axis) and energy difference (y-axis) across the four evaluated models. Each point is a concept, with size proportional to activation frequency.

394 We next examine whether conceptual blindspots arise only from model architecture or also from
 395 distributional properties of the concepts themselves. Specifically, we test whether concepts that
 396 are rarely activated in natural images—the long tail of the data distribution—are more likely to
 397 become blindspots in generative models. We process the natural dataset $\mathcal{D}_X$ with the trained SAE
 398 and compute, for each concept k , its empirical frequency $\|\mathbf{Z}_{:,k}\|_0$, where $\mathbf{Z}_{:,k}$ contains activations
 399 of concept k across all images. We then compare this frequency with the absolute energy difference
 400 observed in generated outputs. Fig. 13 visualizes the relationship for all evaluated models. Concepts
 401 with higher natural-data frequency tend to show lower energy discrepancies, whereas rare concepts—
 402 especially suppressed ones ($\delta(k) < 0.5$)—show larger alignment errors. This pattern suggests that
 403 many blindspots are linked to long-tail concept distributions rather than only random model quirks.
 404 Mitigating them may require data-centric strategies, such as reweighting or augmentation, in addition
 405 to architectural changes.

6 Discussion

Our analysis identifies multiple conceptual blindspots across four widely used generative image models, while also showing that the same measurement framework can support several levels of inquiry. Rather than treating each finding as an isolated case study, we use the method and exploratory tool to demonstrate a broader diagnostic workflow: identifying concepts that models struggle to generate, detecting memorization artifacts, finding datapoints with insufficient captions, quantifying post-training effects, and characterizing conceptual shifts across model architectures. These results leave substantial room for follow-up work that expands the evaluated model classes, studies additional datasets and domains, and investigates the root causes of conceptual blindspots. Future work could also explore hierarchical RA-SAE representations to support more nuanced analysis of related concepts.

Limitations. Several limitations bound the scope of our conclusions. Because concept extraction relies on DINOv2 and RA-SAE, the analysis is limited to concepts represented by these models; concepts that are poorly encoded by the feature backbone or sparse dictionary may remain invisible. In addition, although the evaluation uses 10,000 images, this sample may still undersample rare concepts, concept co-occurrences, and other compositional statistics in the long tail.

References

- [1] OpenAI. Sora: Creating video from text, 2024. URL <https://openai.com/sora/>
- [2] William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4195–4205, 2023.
- [3] Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever. Zero-shot text-to-image generation. In *International conference on machine learning*, pages 8821–8831. PMLR, 2021.
- [4] Chitwan Saharia, William Chan, Saurabh Saxena, Lala Li, Jay Whang, Emily L Denton, Kamyar Ghasemipour, Raphael Gontijo Lopes, Burcu Karagol Ayan, Tim Salimans, et al. Photorealistic text-to-image diffusion models with deep language understanding. *Advances in neural information processing systems*, 35:36479–36494, 2022.
- [5] Alex Nichol, Prafulla Dhariwal, Aditya Ramesh, Pranav Shyam, Pamela Mishkin, Bob McGrew, Ilya Sutskever, and Mark Chen. GLIDE: Towards photorealistic image generation and editing with text-guided diffusion models. *arXiv:2112.10741*, 2021.
- [6] Chen Wang, Jiatao Gu, Xiaoxiao Long, Yuan Liu, and Lingjie Liu. Geco: Generative image-to-3d within a second. *arXiv preprint arXiv:2405.20327*, 2024.
- [7] Ben Poole, Ajay Jain, Jonathan T Barron, and Ben Mildenhall. Dreamfusion: Text-to-3d using 2d diffusion. *arXiv preprint arXiv:2209.14988*, 2022.
- [8] Elad Richardson, Gal Metzer, Yuval Alaluf, Raja Giryes, and Daniel Cohen-Or. Texture: Text-guided texturing of 3d shapes. In *ACM SIGGRAPH 2023 conference proceedings*, pages 1–11, 2023.
- [9] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10684–10695, 2022.
- [10] Michael Ahn, Anthony Brohan, Noah Brown, Yevgen Chebotar, Omar Cortes, Byron David, Chelsea Finn, Keerthana Gopalakrishnan, Karol Hausman, Alex Herzog, et al. Do as i can, not as i say: Grounding language in robotic affordances. *arXiv preprint arXiv:2204.01691*, 2022.
- [11] Wenlong Huang, Fei Xia, Ted Xiao, Harris Chan, Jacky Liang, Pete Florence, Andy Zeng, Jonathan Tompson, Igor Mordatch, Yevgen Chebotar, et al. Inner monologue: Embodied reasoning through planning with language models. *arXiv preprint arXiv:2207.05608*, 2022.
- [12] Wenlong Huang, Pieter Abbeel, Deepak Pathak, and Igor Mordatch. Language models as zero-shot planners: Extracting actionable knowledge for embodied agents. In *International Conference on Machine Learning*, pages 9118–9147. PMLR, 2022.
- [13] Yun-Chun Chen, Selena Ling, Zhiqin Chen, Vladimir G Kim, Matheus Gadelha, and Alec Jacobson. Text-guided controllable mesh refinement for interactive 3d modeling. In *SIGGRAPH Asia 2024 Conference Papers*, pages 1–11, 2024.
- [14] Lei Zhong, Yiming Xie, Varun Jampani, Deqing Sun, and Huaizu Jiang. Smoodi: Stylized motion diffusion model. In *European Conference on Computer Vision*, pages 405–421. Springer, 2024.
- [15] Yawar Siddiqui, Antonio Alliegro, Alexey Artemov, Tatiana Tommasi, Daniele Sirigatti, Vladislav Rosov, Angela Dai, and Matthias Nießner. Meshgpt: Generating triangle meshes with decoder-only transformers. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 19615–19625, 2024.
- [16] Gary Marcus, Ernest Davis, and Scott Aaronson. A very preliminary analysis of dall-e 2. *arXiv preprint arXiv:2204.13807*, 2022.
- [17] Ángel Alexander Cabrera, Abraham J Druck, Jason I Hong, and Adam Perer. Discovering and validating ai errors with crowdsourced failure reports. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW2):1–22, 2021.

- [18] Rebecca Heigl. Generative artificial intelligence in creative contexts: a systematic review and future research agenda. *Management Review Quarterly*, pages 1–38, 2025.
- [19] Qihao Liu, Adam Kortylewski, Yutong Bai, Song Bai, and Alan Yuille. Discovering failure modes of text-guided diffusion models via adversarial search. *arXiv preprint arXiv:2306.00974*, 2023.
- [20] Colin Conwell, Rupert Tawiah-Quashie, and Tomer Ullman. Relations, negations, and numbers: Looking for logic in generative text-to-image models. *arXiv preprint arXiv:2411.17066*, 2024.
- [21] Wenquan Lu, Yufei Xu, Jing Zhang, Chaoyue Wang, and Dacheng Tao. HandRefiner: Refining malformed hands in generated images by diffusion-based conditional inpainting. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pages 7085–7093, 2024.
- [22] Supreeth Narasimhaswamy, Uttaran Bhattacharya, Xiang Chen, Ishita Dasgupta, Saayan Mitra, and Minh Hoai. HanDiffuser: Text-to-image generation with realistic hand appearances. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2468–2479, 2024.
- [23] Qilong Zhangli, Jindong Jiang, Di Liu, Licheng Yu, Xiaoliang Dai, Ankit Ramchandani, Guan Pang, Dimitris N Metaxas, and Praveen Krishnan. Layout-agnostic scene text image synthesis with diffusion models. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7496–7506. IEEE Computer Society, 2024.
- [24] Forouzan Fallah, Maitreya Patel, Agneet Chatterjee, Vlad I Morariu, Chitta Baral, and Yezhou Yang. TextInVision: Text and prompt complexity driven visual text generation benchmark. *arXiv:2503.13730*, 2025.
- [25] Yuefan Cao, Xuyang Guo, Jiayan Huo, Yingyu Liang, Zhenmei Shi, Zhao Song, Jiahao Zhang, and Zhen Zhuang. Text-to-image diffusion models cannot count, and prompt refinement cannot help. *arXiv preprint arXiv:2503.06884*, 2025.
- [26] Colin Conwell and Tomer Ullman. Testing relational understanding in text-guided image generation. *arXiv preprint arXiv:2208.00005*, 2022.
- [27] Mahzarin R Banaji and Anthony G Greenwald. *Blindspot: Hidden biases of good people*. Bantam, 2016.
- [28] George Stein, Jesse Cresswell, Rasa Hosseinzadeh, Yi Sui, Brendan Ross, Valentin Vilecroze, Zhaoyan Liu, Anthony L Caterini, Eric Taylor, and Gabriel Loaiza-Ganem. Exposing flaws of generative model evaluation metrics and their unfair treatment of diffusion models. *Advances in Neural Information Processing Systems*, 36:3732–3784, 2023.
- [29] Boheng Wang, Yunhuai Zhu, Liuqing Chen, Jingcheng Liu, Lingyun Sun, and Peter Childs. A study of the evaluation metrics for generative images containing combinational creativity. *AI EDAM*, 37:e11, 2023.
- [30] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems*, 30, 2017.
- [31] Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. Clipscore: A reference-free evaluation metric for image captioning. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 7514–7528, 2021.
- [32] Mischa Dombrowski, Weitong Zhang, Sarah Cechnicka, Hadrien Reynaud, and Bernhard Kainz. Image generation diversity issues and how to tame them. *arXiv:2411.16171*, 2024.
- [33] Jaehui Hwang, Junghyuk Lee, and Jong-Seok Lee. Anomaly score: Evaluating generative models and individual generated images based on complexity and vulnerability. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8754–8763, 2024.

- [34] Lucas Theis, Aäron van den Oord, and Matthias Bethge. A note on the evaluation of generative models. *arXiv:1511.01844*, 2015.
- [35] Muhammad Ferjad Naeem, Seong Joon Oh, Youngjung Uh, Yunjey Choi, and Jaejun Yoo. Reliable fidelity and diversity metrics for generative models. In *International conference on machine learning*, pages 7176–7185. PMLR, 2020.
- [36] Vitali Petsiuk, Alexander E Siemenn, Saisamrit Surbehera, Zad Chin, Keith Tyser, Gregory Hunter, Arvind Raghavan, Yann Hicke, Bryan A Plummer, Ori Kerret, et al. Human evaluation of text-to-image models on a multi-task benchmark. *arXiv:2211.12112*, 2022.
- [37] Jiazheng Xu, Xiao Liu, Yuchen Wu, Yuxuan Tong, Qinkai Li, Ming Ding, Jie Tang, and Yuxiao Dong. Imagereward: Learning and evaluating human preferences for text-to-image generation. *Advances in Neural Information Processing Systems*, 36:15903–15935, 2023.
- [38] Xiaoshi Wu, Yiming Hao, Keqiang Sun, Yixiong Chen, Feng Zhu, Rui Zhao, and Hongsheng Li. Human preference score v2: A solid benchmark for evaluating human preferences of text-to-image synthesis. 2023.
- [39] David Bau, Jun-Yan Zhu, Jonas Wulff, William Peebles, Hendrik Strobelt, Bolei Zhou, and Antonio Torralba. Seeing what a GAN cannot generate. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4502–4511, 2019.
- [40] Karen Simonyan, Andrea Vedaldi, and Andrew Zisserman. Deep inside convolutional networks: Visualising image classification models and saliency maps. *Proceedings of the International Conference on Learning Representations (ICLR)*, 2013.
- [41] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. *Proceedings of the International Conference on Machine Learning (ICML)*, 2017.
- [42] Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-cam: Visual explanations from deep networks via gradient-based localization. *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017.
- [43] Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. *Advances in Neural Information Processing Systems (NIPS)*, 2018.
- [44] Amirata Ghorbani, Abubakar Abid, and James Zou. Interpretation of neural networks is fragile. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2017.
- [45] Peter Hase and Mohit Bansal. Evaluating explainable ai: Which algorithmic explanations help users predict model behavior? *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.
- [46] Been Kim, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viegas, et al. Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (tcav). *Proceedings of the International Conference on Machine Learning (ICML)*, 2018.
- [47] David Bau, Bolei Zhou, Aditya Khosla, Aude Oliva, and Antonio Torralba. Network dissection: Quantifying interpretability of deep visual representations. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
- [48] Thomas Fel, Agustin Picard, Louis Bethune, Thibaut Boissin, David Vigouroux, Julien Colin, Rémi Cadène, and Thomas Serre. Craft: Concept recursive activation factorization for explainability. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.
- [49] Matthew Kowal, Achal Dave, Rares Ambrus, Adrien Gaidon, Konstantinos G Derpanis, and Pavel Tokmakov. Understanding video transformers via universal concept discovery. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

- [50] Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. *ArXiv e-print*, 2023.
- [51] Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023.
- [52] Thomas Fel, Victor Boutin, Mazda Moayeri, Remi Cadene, Louis Bethune, Mathieu Chalvidal, and Thomas Serre. A holistic approach to unifying automatic concept extraction and concept importance estimation. *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [53] David Chanin, James Wilken-Smith, Tomas Dulka, Hardik Bhatnagar, and Joseph Bloom. A is for absorption: Studying feature splitting and absorption in sparse autoencoders. *ArXiv e-print*, 2024.
- [54] Martin Wattenberg and Fernanda B Viegas. Relational composition in neural networks: A survey and call to action. *ArXiv e-print*, 2024.
- [55] Usha Bhalla, Suraj Srinivas, Asma Ghandeharioun, and Himabindu Lakkaraju. Towards unifying interpretability and control: Evaluation via intervention. *ArXiv e-print*, 2024.
- [56] Thomas Fel, Ekdeep Singh Lubana, Jacob S Prince, Matthew Kowal, Victor Boutin, Isabel Papadimitriou, Binxu Wang, Martin Wattenberg, Demba Ba, and Talia Konkle. Archetypal SAE: Adaptive and stable dictionary learning for concept extraction in large vision models. 2025.
- [57] Arpita Chowdhury, Dipanjyoti Paul, Zheda Mai, Jianyang Gu, Ziheng Zhang, Kazi Sajeed Mehrab, Elizabeth G Campolongo, Daniel Rubenstein, Charles V Stewart, Anuj Karpatne, et al. Prompt-CAM: A simpler interpretable transformer for fine-grained analysis. *arXiv:2501.09333*, 2025.
- [58] David Bau, Jun-Yan Zhu, Hendrik Strobelt, Bolei Zhou, Joshua B Tenenbaum, William T Freeman, and Antonio Torralba. GAN dissection: Visualizing and understanding generative adversarial networks. 2018.
- [59] Raphael Tang, Linqing Liu, Akshat Pandey, Zhiying Jiang, Gefei Yang, Karun Kumar, Pontus Stenetorp, Jimmy Lin, and Ferhan Ture. What the DAAM: Interpreting stable diffusion using cross attention. 2022.
- [60] Ali Borji. Qualitative failures of image generation models and their application in detecting deepfakes. *Image and Vision Computing*, 137:104771, 2023.
- [61] Aditya Ramesh, Prafulla Dhariwal, Alex Nichol, Casey Chu, and Mark Chen. Hierarchical text-conditional image generation with CLIP latents. *arXiv preprint arXiv:2204.06125*, 1(2):3, 2022.
- [62] Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. *arXiv preprint arXiv:2011.13456*, 2020.
- [63] Yang Song, Prafulla Dhariwal, Mark Chen, and Ilya Sutskever. Consistency models. 2023.
- [64] Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *International conference on machine learning*, pages 8162–8171. PMLR, 2021.
- [65] Jonathan Ho, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi, and David J Fleet. Video diffusion models. *Advances in Neural Information Processing Systems*, 35:8633–8646, 2022.
- [66] Haoyu Lu, Guoxing Yang, Nanyi Fei, Yuqi Huo, Zhiwu Lu, Ping Luo, and Mingyu Ding. VDT: General-purpose video diffusion transformers via mask modeling. *arXiv preprint arXiv:2305.13311*, 2023.

- [67] Jiuniu Wang, Hangjie Yuan, Dayou Chen, Yingya Zhang, Xiang Wang, and Shiwei Zhang. ModelScope text-to-video technical report. *arXiv preprint arXiv:2308.06571*, 2023.
- [68] Yuwei Guo, Ceyuan Yang, Anyi Rao, Zhengyang Liang, Yaohui Wang, Yu Qiao, Maneesh Agrawala, Dahua Lin, and Bo Dai. AnimateDiff: Animate your personalized text-to-image diffusion models without specific tuning. *arXiv preprint arXiv:2307.04725*, 2023.
- [69] Shanchuan Lin and Xiao Yang. AnimateDiff-Lightning: Cross-model diffusion distillation. *arXiv preprint arXiv:2403.12706*, 2024.
- [70] Wenyi Hong, Ming Ding, Wendi Zheng, Xinghan Liu, and Jie Tang. CogVideo: Large-scale pretraining for text-to-video generation via transformers. *arXiv preprint arXiv:2205.15868*, 2022.
- [71] Haoxin Chen, Menghan Xia, Yingqing He, Yong Zhang, Xiaodong Cun, Shaoshu Yang, Jinbo Xing, Yaofang Liu, Qifeng Chen, Xintao Wang, et al. VideoCrafter1: Open diffusion models for high-quality video generation. *arXiv preprint arXiv:2310.19512*, 2023.
- [72] Jay Zhangjie Wu, Yixiao Ge, Xintao Wang, Stan Weixian Lei, Yuchao Gu, Yufei Shi, Wynne Hsu, Ying Shan, Xiaohu Qie, and Mike Zheng Shou. Tune-a-video: One-shot tuning of image diffusion models for text-to-video generation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 7623–7633, 2023.
- [73] Chen-Hsuan Lin, Jun Gao, Luming Tang, Towaki Takikawa, Xiaohui Zeng, Xun Huang, Karsten Kreis, Sanja Fidler, Ming-Yu Liu, and Tsung-Yi Lin. Magic3D: High-resolution text-to-3d content creation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 300–309, 2023.
- [74] Heewoo Jun and Alex Nichol. Shap-E: Generating conditional 3d implicit functions. *arXiv preprint arXiv:2305.02463*, 2023.
- [75] Haochen Wang, Xiaodan Du, Jiahao Li, Raymond A Yeh, and Greg Shakhnarovich. Score jacobian chaining: Lifting pretrained 2d diffusion models for 3d generation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 12619–12629, 2023.
- [76] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020.
- [77] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502*, 2020.
- [78] Stability AI. Stable Diffusion 2.0 Release, 2022. URL <https://stability.ai/news/stable-diffusion-v2-release>
- [79] Stability AI. Introducing Stable Diffusion 3, 2024. URL <https://stability.ai/news/stable-diffusion-3>
- [80] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. SDXL: Improving latent diffusion models for high-resolution image synthesis. *arXiv preprint arXiv:2307.01952*, 2023.
- [81] Kihyuk Sohn, Nataniel Ruiz, Kimin Lee, Daniel Castro Chin, Irina Blok, Huiwen Chang, Jarred Barber, Lu Jiang, Glenn Entis, Yuanzhen Li, et al. StyleDrop: Text-to-image generation in any style. *arXiv preprint arXiv:2306.00983*, 2023.
- [82] Zhihong Pan, Xin Zhou, and Hao Tian. Arbitrary style guidance for enhanced diffusion-based text-to-image generation. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pages 4461–4471, 2023.
- [83] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 3836–3847, 2023.

- [84] Nataniel Ruiz, Yuanzhen Li, Varun Jampani, Yael Pritch, Michael Rubinstein, and Kfir Aberman. DreamBooth: Fine tuning text-to-image diffusion models for subject-driven generation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 22500–22510, 2023.
- [85] Darian Tomavsevic, Fadi Boutros, Chenhao Lin, Naser Damer, Vitomir Struc, and Peter Peer. Id-booth: Identity-consistent face generation with diffusion models. *arXiv preprint arXiv:2504.07392*, 2025.
- [86] Anton Razzhigaev, Arseniy Shakhmatov, Anastasia Maltseva, Vladimir Arkhipkin, Igor Pavlov, Ilya Ryabov, Angelina Kuts, Alexander Panchenko, Andrey Kuznetsov, and Denis Dimitrov. Kandinsky: An improved text-to-image synthesis with image prior and latent diffusion. *arXiv preprint arXiv:2310.03502*, 2023.
- [87] Junsong Chen, Jincheng Yu, Chongjian Ge, Lewei Yao, Enze Xie, Yue Wu, Zhongdao Wang, James Kwok, Ping Luo, Huchuan Lu, et al. Pixart- α : Fast training of diffusion transformer for photorealistic text-to-image synthesis. *arXiv preprint arXiv:2310.00426*, 2023.
- [88] Stability AI. DeepFloyd IF: A Powerful Open-Source Text-to-Image Model, 2023. URL <https://stability.ai/news/deepfloyd-if-text-to-image-model>.
- [89] Black Forest Labs. Announcing Black Forest Labs, 2024. URL <https://blackforestlabs.ai/announcing-black-forest-labs/>.
- [90] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *Proceedings of the International Conference on Learning Representations (ICLR)*, 2020.
- [91] Jiahui Yu, Yuanzhong Xu, Jing Yu Koh, Thang Luong, Gunjan Baid, Zirui Wang, Vijay Vasudevan, Alexander Ku, Yinfei Yang, Burcu Karagol Ayan, et al. Scaling autoregressive models for content-rich text-to-image generation. *arXiv preprint arXiv:2206.10789*, 2(3):5, 2022.
- [92] Joel Hestness, Sharan Narang, Newsha Ardalani, Gregory Diamos, Heewoo Jun, Hassan Kianinejad, Md Mostofa Ali Patwary, Yang Yang, and Yanqi Zhou. Deep learning scaling is predictable, empirically. *arXiv preprint arXiv:1712.00409*, 2017.
- [93] Christoph Schuhmann, Romain Beaumont, Richard Vencu, Cade Gordon, Ross Wightman, Mehdi Cherti, Theo Coombes, Aarush Katta, Clayton Mullis, Mitchell Wortsman, et al. LAION-5B: An open large-scale dataset for training next generation image-text models. *Advances in neural information processing systems*, 35:25278–25294, 2022.
- [94] Common Crawl. Common crawl corpus. <https://commoncrawl.org> 2017.
- [95] Minwoo Byeon, Beomhee Park, Haecheon Kim, Sungjun Lee, Woonhyuk Baek, and Saehoon Kim. COYO-700M: Image-text pair dataset, 2022.
- [96] Piyush Sharma, Nan Ding, Sebastian Goodman, and Radu Soricut. Conceptual Captions: A cleaned, hypernymed, image alt-text dataset for automatic image captioning. In *Proceedings of ACL*, 2018.
- [97] Abeba Birhane, Sanghyun Han, Vishnu Boddeti, Sasha Luccioni, et al. Into the LAION’s den: Investigating hate in multimodal datasets. *Advances in neural information processing systems*, 36:21268–21284, 2023.
- [98] Abeba Birhane, Sepehr Dehdashtian, Vinay Prabhu, and Vishnu Boddeti. The dark side of dataset scaling: Evaluating racial classification in multimodal models. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency*, pages 1229–1244, 2024.
- [99] Preethi Seshadri, Sameer Singh, and Yanai Elazar. The bias amplification paradox in text-to-image generation. *arXiv preprint arXiv:2308.00755*, 2023.

- 711 [100] Abeba Birhane, Vinay Uday Prabhu, and Emmanuel Kahembwe. Multimodal datasets: Misog-
712 ny, pornography, and malignant stereotypes. *arXiv preprint arXiv:2110.01963*, 2021.
- 713 [101] David Thiel. Identifying and eliminating CSAM in generative ml training data and models.
714 *Stanford Internet Observatory, Cyber Policy Center, December*, 23:3, 2023.
- 715 [102] Ali Shirali and Moritz Hardt. What makes ImageNet look unlike LAION. *arXiv preprint*
716 *arXiv:2306.15769*, 2023.
- 717 [103] Thao Nguyen, Samir Yitzhak Gadre, Gabriel Ilharco, Sewoong Oh, and Ludwig Schmidt.
718 Improving multimodal datasets with image captioning. *Advances in Neural Information*
719 *Processing Systems*, 36:22047–22069, 2023.
- 720 [104] Bruno A Olshausen and David J Field. Emergence of simple-cell receptive field properties by
721 learning a sparse code for natural images. *Nature*, 1996.
- 722 [105] Michael Elad. Sparse and redundant representations: from theory to applications in signal and
723 image processing. 2010.
- 724 [106] Julien Mairal, Francis Bach, and Jean Ponce. Sparse modeling for image and vision processing.
725 *Foundations and Trends in Computer Graphics and Vision*, 2014.
- 726 [107] David L Donoho. Compressed sensing. *IEEE Transactions on Information Theory*, 2006.
- 727 [108] Emmanuel J Candès, Justin Romberg, and Terence Tao. Robust uncertainty principles: Exact
728 signal reconstruction from highly incomplete frequency information. *IEEE Transactions on*
729 *Information Theory*, 2006.
- 730 [109] Stuart Lloyd. Least squares quantization in pcm. *IEEE Transactions on Information Theory*,
731 1982.
- 732 [110] Daniel D Lee and H Sebastian Seung. Learning the parts of objects by non-negative matrix
733 factorization. *Nature*, 1999.
- 734 [111] Nicolas Gillis. Nonnegative matrix factorization. 2020.
- 735 [112] Hui Zou, Trevor Hastie, and Robert Tibshirani. Sparse principal component analysis. *Journal*
736 *of Computational and Graphical Statistics*, 2006.
- 737 [113] Michal Aharon, Michael Elad, and Alfred Bruckstein. K-svd: An algorithm for designing
738 overcomplete dictionaries for sparse representation. *IEEE Transactions on Signal Processing*,
739 2006.
- 740 [114] Julien Mairal, Francis Bach, Jean Ponce, and Guillermo Sapiro. Online dictionary learning for
741 sparse coding. *Proceedings of the International Conference on Machine Learning (ICML)*,
742 2009.
- 743 [115] Rodolphe Jenatton, Guillaume Obozinski, and Francis Bach. Structured sparse principal
744 component analysis. *International Conference on Artificial Intelligence and Statistics*, 2010.
- 745 [116] Daniel A Spielman, Huan Wang, and John Wright. Exact recovery of sparsely-used dictionaries.
746 *The Journal of Machine Learning Research (JMLR)*, 2012.
- 747 [117] Jean Barbier and Nicolas Macris. Statistical limits of dictionary learning: random matrix
748 theory and the spectral replica method. *Physical Review E*, 2022.
- 749 [118] Vardan Papyan, Yaniv Romano, and Michael Elad. Convolutional dictionary learning via local
750 processing. *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*,
751 2017.
- 752 [119] Alex Tamkin, Mohammad Tafseeque, and Noah D Goodman. Codebook features: Sparse and
753 discrete interpretability for neural networks. *ArXiv e-print*, 2023.
- 754 [120] Berk Tinaz, Zalan Fabian, and Mahdi Soltanolkotabi. Emergence and evolution of interpretable
755 concepts in diffusion models. *arXiv:2504.15473*, 2025.

- [121] Akshay Kulkarni, Ge Yan, Chung-En Sun, Tuomas Oikarinen, and Tsui-Wei Weng. Interpretable generative models through post-hoc concept bottlenecks. *arXiv preprint arXiv:2503.19377*, 2025.
- [122] Mert Yuksekgonul, Maggie Wang, and James Zou. Post-hoc concept bottleneck models. *arXiv preprint arXiv:2205.15480*, 2022.
- [123] Tuomas Kynkäänniemi, Tero Karras, Samuli Laine, Jaakko Lehtinen, and Timo Aila. Improved precision and recall metric for assessing generative models. *Advances in neural information processing systems*, 32, 2019.
- [124] Jingyi Xu, Hieu Le, and Dimitris Samaras. Assessing sample quality via the latent space of generative models. In *European Conference on Computer Vision*, pages 449–464. Springer, 2024.
- [125] Iro Laina, Yuki M Asano, and Andrea Vedaldi. Measuring the interpretability of unsupervised representations via quantized reverse probing. *arXiv:2209.03268*, 2022.
- [126] Julius Von Kugelgen, Yash Sharma, Luigi Gresele, Wieland Brendel, Bernhard Scholkopf, Michel Besserve, and Francesco Locatello. Self-supervised learning with data augmentations provably isolates content from style. *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [127] Francesco Locatello, Stefan Bauer, Mario Lucic, Gunnar Raetsch, Sylvain Gelly, Bernhard Scholkopf, and Olivier Bachem. Challenging common assumptions in the unsupervised learning of disentangled representations. *Proceedings of the International Conference on Machine Learning (ICML)*, 2019.
- [128] Roland S Zimmermann, Yash Sharma, Steffen Schneider, Matthias Bethge, and Wieland Brendel. Contrastive learning inverts the data generating process. *Proceedings of the International Conference on Machine Learning (ICML)*, 2021.
- [129] Luigi Gresele, Paul K Rubenstein, Arash Mehrjou, Francesco Locatello, and Bernhard Scholkopf. The incomplete rosetta stone problem: Identifiability results for multi-view nonlinear ica. *Uncertainty in Artificial Intelligence*, 2020.
- [130] Luigi Gresele, Julius Von Kugelgen, Vincent Stimper, Bernhard Scholkopf, and Michel Besserve. Independent mechanism analysis, a new concept? *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [131] Maya Okawa, Ekdeep S Lubana, Robert Dick, and Hidenori Tanaka. Compositional abilities emerge multiplicatively: Exploring diffusion models on a synthetic task. *Advances in Neural Information Processing Systems*, 36:50173–50195, 2023.
- [132] Core Francisco Park, Maya Okawa, Andrew Lee, Ekdeep S Lubana, and Hidenori Tanaka. Emergence of hidden capabilities: Exploring learning dynamics in concept space. *Advances in Neural Information Processing Systems*, 37:84698–84729, 2024.
- [133] Ilyes Khemakhem, Diederik Kingma, Ricardo Monti, and Aapo Hyvarinen. Variational autoencoders and nonlinear ica: A unifying framework. *Proceedings of the International Conference on Machine Learning (ICML)*, 2020.
- [134] Aapo Hyvarinen, Hiroaki Sasaki, and Richard Turner. Nonlinear ICA using auxiliary variables and generalized contrastive learning. In *The 22nd International Conference on Artificial Intelligence and Statistics*, pages 859–868. PMLR, 2019.
- [135] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *ArXiv e-print*, 2023.
- [136] Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, Roger Grosse, Sam McCandlish, Jared Kaplan, Dario Amodei, Martin Wattenberg, and Christopher Olah. Toy models of superposition. *Transformer Circuits Thread*, 2022.

- 805 [137] Leo Gao, Tom Dupre la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya
806 Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders. *Proceedings*
807 *of the International Conference on Learning Representations (ICLR)*, 2025.
- 808 [138] Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, Brian Chen,
809 Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jones, Hoagy Cunningham, Nicholas L
810 Turner, Callum McDougall, Monte MacDiarmid, C. Daniel Freeman, Theodore R. Sumers,
811 Edward Rees, Joshua Batson, Adam Jermy, Shan Carter, Chris Olah, and Tom Henighan.
812 Scaling monosemanticity: Extracting interpretable features from claude 3 sonnet. *Transformer*
813 *Circuits Thread*, 2024.
- 814 [139] Senthooan Rajamanoharan, Tom Lieberum, Nicolas Sonnerat, Arthur Conmy, Vikrant Varma,
815 Janos Kramar, and Neel Nanda. Jumping ahead: Improving reconstruction fidelity with
816 jumprelu sparse autoencoders. *ArXiv e-print*, 2024.
- 817 [140] Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward
818 Grefenstette, and Roberta Raileanu. Understanding the effects of rlhf on llm generalisation
819 and diversity. *arXiv preprint arXiv:2310.06452*, 2023.
- 820 [141] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and
821 Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model.
822 *Advances in Neural Information Processing Systems*, 36:53728–53741, 2023.
- 823 [142] Colin McDiarmid et al. On the method of bounded differences. *Surveys in combinatorics*, 141
824 (1):148–188, 1989.

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Appendix

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A Experimental Setup

This section details the experimental setup used to compare four widely used generative image models: SD 1.5/2.1, Kandinsky, and PixArt, all trained on LAION-5B or related subsets and derivatives. The code is available at a software repository.

A.1 Observation Space

The observation space is constructed from 10,000 image-text pairs sampled from LAION-5B [93], which provides the natural-image reference domain for the experiments. Because the original release is no longer available due to CSAM and other unsafe-content concerns, we use a filtered substitute release at <https://huggingface.co/datasets/laion/relaion2B-en-research-safe>

The sampling procedure consists of: (1) loading LAION with the Hugging Face `datasets` library, (2) validating URL structure and resource availability through HTTP HEAD requests, and (3) randomly sampling with replacement until 10,000 valid samples are obtained. This produces D_G as accessible (x, t) tuples, enabling controlled comparison despite the partial unavailability of the original image URLs. Additional synthesized examples appear in Appendix H

A.2 Synthesized Images

For each evaluated model, we generate a synthetic dataset D_{g_θ} in one-to-one correspondence with D_G , yielding triplets (x, x', t) . For every one of the 10,000 image-text pairs (x, t) , the caption t is used to synthesize a counterpart image x' with model g_θ .

The synthesis process follows the standard text-to-image pipeline for each architecture, implemented with the Hugging Face `diffusers` library and mixed-precision model loading (fp16). All synthetic images are generated at 512×512 resolution with default parameters.

Stable Diffusion 1.5. The checkpoint from <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5> (which is a mirror of the deprecated <https://huggingface.co/runwayml/stable-diffusion-v1-5>) is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Stable Diffusion 1.5 + DPO. The DPO variant of SD 1.5 (used in the analysis in Sec. 5.5) follows the baseline SD 1.5 implementation, but replaces the UNet component with a DPO-trained version from <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>.

Stable Diffusion 2.1. The checkpoint from <https://huggingface.co/stabilityai/stable-diffusion-2-1> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Kandinsky. The checkpoint from <https://huggingface.co/kandinsky-community/kandinsky-2-1> is used. Inference is performed using 100 inference steps, with the guidance scale fixed at 4.0.

PixArt. The checkpoint from <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

A.3 Distribution Level Analysis

Section 5.1 We compute energy differences $\delta(\cdot)$ over all 32,000 concepts for each evaluated model. A sigmoid transformation with temperature $T = 0.8$ is applied during normalization, and the resulting values are visualized as log-scale density histograms with 100 bins spanning $[0, 1]$.

Section 5.2 We embed all 32,000 concepts into two dimensions using UMAP on the sparse concept codes. Each point represents one concept and is colored by its energy difference $\delta(\cdot)$, highlighting both suppressed and exaggerated blindspots. To quantify cross-model consistency, we compute pairwise Pearson correlations between $\delta(\cdot)$ vectors for all model pairs and visualize them with scatter

plots and correlation matrices. This analysis shows whether blindspots cluster in concept space and distinguishes model-specific from shared blindspot patterns.

Section 5.3 We rank all 32,000 concepts by $\delta(\cdot)$ and manually inspect the extremes, covering both suppressed and exaggerated blindspots. Suppressed blindspots use the threshold $\delta(\cdot) < 0.1$, and exaggerated blindspots use $\delta(\cdot) > 0.9$. The presented examples are manually annotated by inspecting the most activating images and spatial attention patterns. Appendix D discusses efforts to automate this concept interpretation.

Section 5.5 We compare SD 1.5 variants with and without DPO. For each image pair (x, x') , we compute the L2 norm between their concept energy vectors, $\|\xi(x') - \xi(x)\|_2$. A sigmoid transformation with temperature $T = 0.8$ is applied to element-wise differences before averaging, producing datapoint-wise energy differences that measure concept-space deviation from the natural counterpart. These differences are visualized as overlapping histograms for the two variants.

Section 5.6 For each concept c_k , we count its empirical frequency $\|\mathbf{Z}_{:,i}\|_0$, defined as the number of non-zero activations across the natural dataset. A sigmoid normalization with temperature $T = 0.4$ is then applied to the energy differences $\delta(\cdot)$. Scatter plots visualize empirical concept frequency on the x-axis and sigmoid-transformed energy difference on the y-axis, with point size proportional to activation frequency and color proportional to energy-difference magnitude.

A.4 • Datapoint Level Analysis

Section 5.4 For each image pair (x, x') , we compute the L2 norm between concept energy vectors, $\|\xi(x') - \xi(x)\|_2$. This gives a scalar measure of conceptual divergence per pair, allowing samples to be ranked by energy difference. Minimal divergence can indicate memorization artifacts, while maximal divergence points to substantial conceptual failures. The ranking enables qualitative inspection of specific failure modes.

A.5 • Co-occurrence Analysis

Appendix F For the natural and synthesized datasets D_G and $D_{g\theta}$, we analyze concept co-occurrence through the co-activation matrix $Z^T Z$, which records pairwise correlations in concept usage. Spectral analysis via eigendecomposition identifies dominant conceptual directions. Alignment between natural and synthetic co-occurrence structures is measured with cosine-similarity heatmaps over the top-100 eigenvectors of each co-occurrence matrix. These 100×100 matrices are visualized as square heatmaps: diagonal alignment indicates shared principal concept axes, while off-diagonal structure reveals rotations and mismatches in compositional geometry.

B Custom RA-SAE Training Details

This section summarizes the training details of the custom archetypal SAE (RA-SAE) [56] used as the concept basis for the paper. The model is open-sourced at a software repository.

Dataset. The auto-encoder is trained on the complete ImageNet-1k training split, containing ≈ 1.28 M RGB images. Each image is converted to 261 visual tokens using DINOv2 [135]; tokens are provided to the SAE without class or position embeddings. The total number of training tokens is therefore $50 \times 1.28\text{M} \times 261 \approx 1.67 \times 10^{10}$.

Dictionary. The dictionary contains 32,000 concept dimensions. Sparse activation uses top- k masking with $k = 5$, setting all activations outside the five largest per input to 0. We initialize weights with Xavier/Glorot and train at mixed precision (fp16), with the final ten epochs run at full precision.

Optimizer and Schedule. The model is trained for 50 epochs with AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$) and weight decay 10^{-5} . Linear warm-up is applied during the first 5% of steps, followed by cosine decay from $\eta_{\max} = 5 \times 10^{-4}$ to $\eta_{\text{final}} = 10^{-6}$. The loss combines MSE with an auxiliary term penalizing activations that never enter the top- k set, using $\lambda = 10^{-5}$.

940 C Computational Resources

941 This section summarizes the GPU resources used for training and evaluation. In total, the experiments
942 used approximately 202 GPU-hours on NVIDIA H100s and H200s.

943 **RA-SAE.** Trained for approximately 24 GPU-hours on three NVIDIA H100s.

944 **Synthesized Images.** Generating the full D_{g_θ} (see Appendix A) required roughly 5 hours per
945 generator when distributed across four NVIDIA H200 GPUs. Across five generators, this totaled
946 approximately 100 GPU-hours on a NVIDIA H200.

947 **● Distribution-Level Analysis.** Extracting energy differences at the distribution level took about 3
948 hours per generator on a single NVIDIA H200 GPU (total ~ 15 GPU-hours).

949 **● Datapoint-Level Analysis.** Computing datapoint-level energy differences, ranking concepts per
950 datapoint, and ranking datapoints per concept also took approximately 3 hours per generator on one
951 H200 GPU (total ~ 15 GPU-hours).

D Autointerpretability

To explore whether RA-SAE concepts can be interpreted automatically, we qualitatively evaluated Vision-Language Models (VLMs) as concept describers. We queried ChatGPT 4o through the OpenAI API with several prompts and image formats for an image x'' in which a given concept was salient, then recorded the VLM-generated description. This section summarizes the findings from that exploration. For evaluation, we allowed the VLM to provide full reasoning; in a scalable deployment, the prompt could instead request only the description and omit the reasoning.

First, we provided the VLM with two inputs: the raw image x'' and a version of x'' with a heatmap superimposed to localize concept intensity:

Input



You are given two images: (1) raw image and (2) the same image with a heatmap of areas that are salient for a concept present in the image, as identified by a DINOv2-based SAE. Describe what is this concept.

Output

The concept identified by the DINOv2-based SAE in these images appears to correspond to **"ring"** or **"metal ring"** as the salient object.

Reasoning:

- In the original image, a sleek, brushed-metal ring is centered on a wooden surface.
- In the heatmap overlay, the regions that are most activated (least transparent with bright color blocks) are tightly focused on the ring itself, particularly the outer surface and inner reflections.
- The wooden background is mostly ignored except for minor scattered activations, confirming that the concept is not about the background texture.
- The consistent and strong focus on the metallic, circular object suggests that the model recognizes this as the defining element—likely trained to detect the shape, material, and reflectivity typical of rings.

In this example, the VLM did not interpret the heatmap correctly. Rather than describing the wood texture in the background, which was the salient RA-SAE region, it described the foreground ring, which received no RA-SAE attention.

To test whether the failure was caused by insufficient prompt detail, we repeated the scenario with an expanded prompt that explained the heatmap more explicitly:

Input



You are given two images: (1) raw image and (2) the same image with a superimposed heatmap of areas that contain a concept identified by a DINOv2-based SAE. Red areas are the most salient (contain the concept); blue areas are the least salient (do not contain the concept). Describe what is this concept.

Output

The concept identified by the DINOv2-based SAE in these images appears to be **"ring"** or more specifically **"metal band/jewelry ring."**

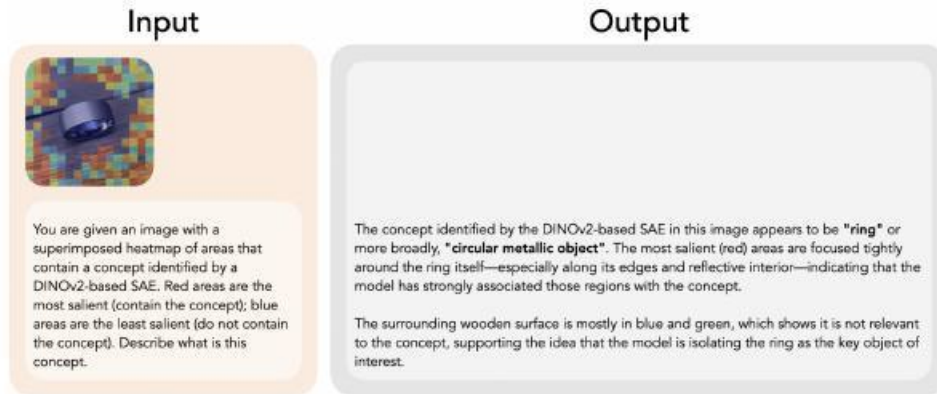
Here's the reasoning:

- In the heatmap image, the red and yellow (most salient) regions align very closely with the circular metal ring at the center of the image.
- The ring is the only distinct object in the photo and is clearly segmented from the background (wooden surface), which is mostly marked in blue and green (least salient).
- The attention does not focus on the texture of the background or lighting but is highly localized to the curved, metallic object.

Thus, the concept is almost certainly the metallic ring—its shape, reflectivity, and material seem to trigger the sparse autoencoder's activation, making it a learned concept in the SAE dictionary.

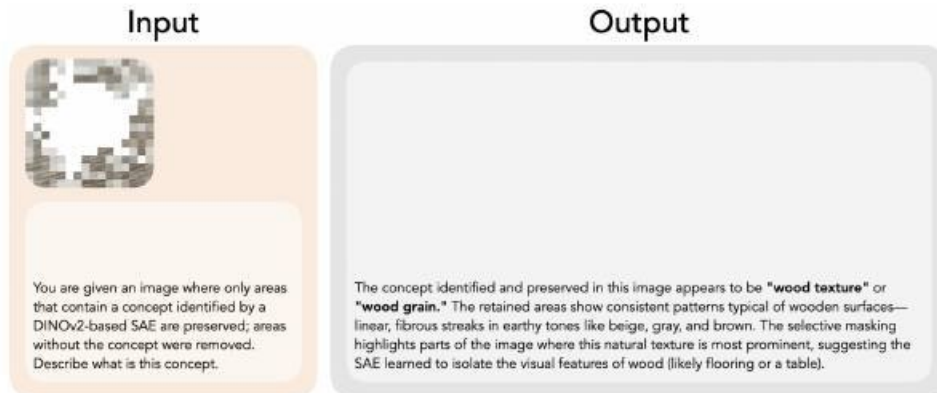
This modification did not resolve the issue: the VLM still identified the ring as the prominent concept. We tried several other prompting strategies, including deeper heatmap explanations and prompts containing the heatmap-generation code, with similar outcomes.

We next provided only the heatmap-superimposed x'' , paired with a similar prompt explaining how to interpret the heatmap:



Again, the VLM failed to identify the wooden background as the prominent concept and instead focused on the ring. Additional prompting strategies with this image format produced largely unchanged results.

These observations suggest that the heatmap alone is not sufficient to direct the VLM toward the salient region; less relevant regions can still distract the model. We therefore replaced the heatmap-superimposed x'' with an alpha-masked version of x'' :



With this format, the VLM correctly identified wood texture as the dominant concept. We observed similar qualitative success across a range of concepts and images.

The alpha-masked formatting of x'' together with the prompt above produced the most reliable results in our qualitative evaluation. The evaluation remains limited by the absence of ground-truth concept annotations and by its focus on a single VLM. Future work on SAE autointerpretability can broaden this analysis across models, prompts, and annotated concept sets.

983 **E Additional Results: Model-Specific Blindspots**

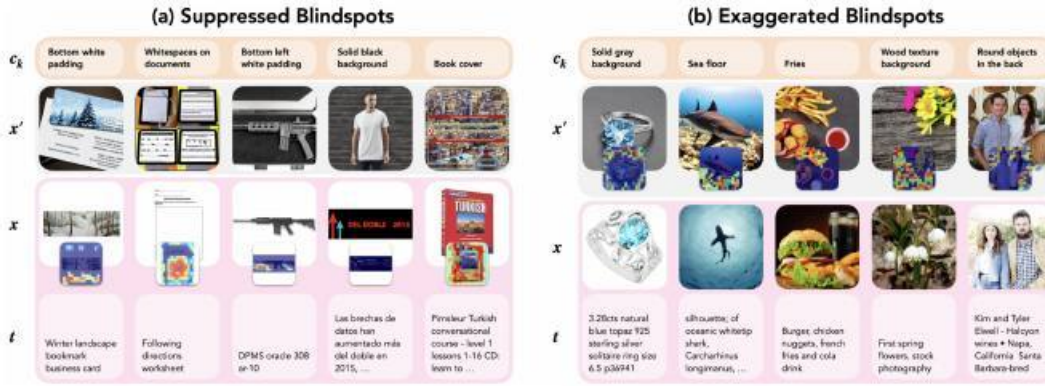


Figure 14: Conceptual blindspot examples for **Stable Diffusion 1.5**. For each concept, the prototypical natural image (for suppressed blindspots) or synthesized image (for exaggerated blindspots) is selected by highest absolute activation, with the concept heatmap superimposed on the image.

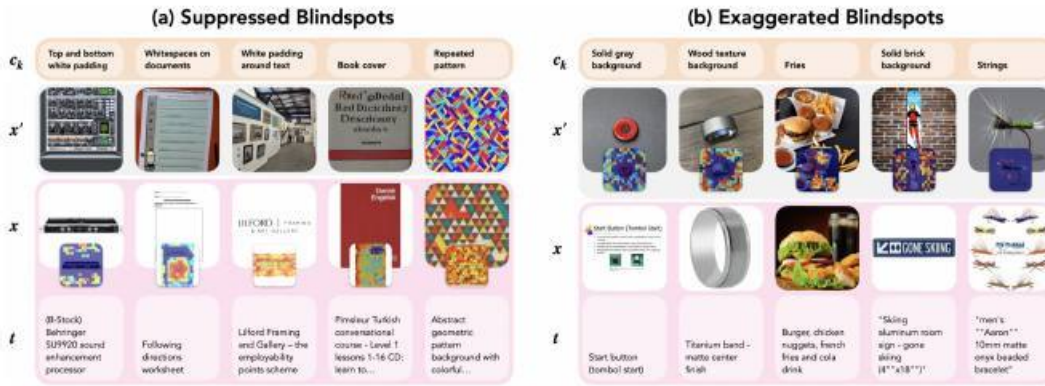


Figure 15: Conceptual blindspot examples for **Stable Diffusion 2.1**. For each concept, the prototypical natural image (for suppressed blindspots) or synthesized image (for exaggerated blindspots) is selected by highest absolute activation, with the concept heatmap superimposed on the image.

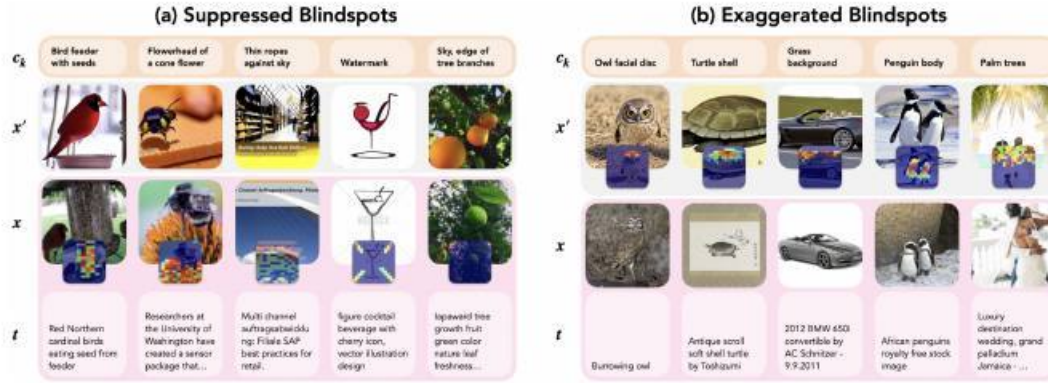


Figure 16: Conceptual blindspot examples for **Kandinsky**. For each concept, the prototypical natural image (for suppressed blindspots) or synthesized image (for exaggerated blindspots) is selected by highest absolute activation, with the concept heatmap superimposed on the image.



Figure 17: Conceptual blindspot examples for **PixArt**. For each concept, the prototypical natural image (for suppressed blindspots) or synthesized image (for exaggerated blindspots) is selected by highest absolute activation, with the concept heatmap superimposed on the image.

F Additional Results: Higher-order Blindspots with Compositional Discrepancy

Thus far, the analysis has focused on individual concept activations. Visual scenes, however, are rarely collections of isolated concepts; they are organized through co-occurrence patterns that encode compositional semantics. We therefore examine whether generative models preserve this higher-order structure by analyzing the co-activation matrix $Z^T Z$, which reflects pairwise correlations in concept usage.

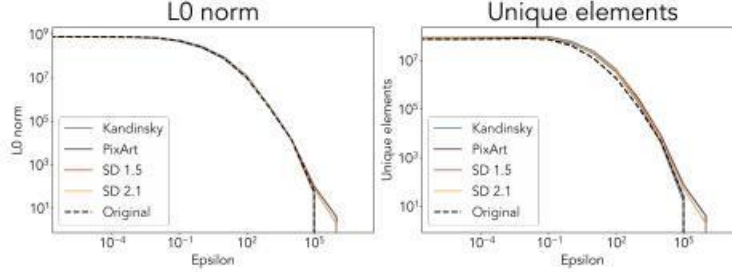


Figure 18: • **Sparsity and Structural Divergence.** Left: L0 norm of the co-occurrence matrix ZZ^T as a function of threshold ϵ , indicating how many entries remain active in each model. Right: number of unique entries in the synthesized distribution relative to the natural distribution. All evaluated models preserve global sparsity structure while diverging in activation content.

At the level of binary structure, diffusion models approximate the global sparsity of the natural co-occurrence matrix with high fidelity. As Fig. 18 (left) shows, the ℓ_0 norm of $Z^T Z$, thresholded at varying ϵ values, tracks closely between natural and synthesized distributions across all models. This indicates that the coarse connectivity of the conceptual graph—which concepts co-activate at all—is broadly preserved. Formally, $Z^T Z$ can be viewed as the adjacency matrix of a weighted, undirected graph over concepts, where edge weights encode co-activation strength across the dataset.

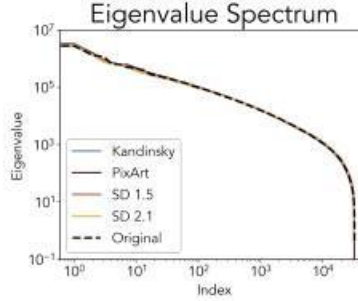


Figure 19: • **Spectral Structure of Co-occurrence.** Log-log plot of eigenvalue spectra from co-occurrence matrices ZZ^T for each model and the natural distribution. All evaluated models match the heavy-tailed decay of the natural distribution.

However, Fig. 18 (right) shows that the specific content of the co-activations diverges: many entries in the model-generated $Z^T Z$ are not shared with the natural baseline. Thus, the capacity for compositional structure may be retained while the identities of active pairings shift, potentially reflecting model-specific inductive biases or training artifacts. To probe these co-occurrence patterns more deeply, we use spectral analysis. Fig. 19 shows eigenvalue spectra for each model’s co-occurrence matrix and for the natural distribution. All spectra have heavy-tailed decay, consistent with power-law behavior, indicating that generative models preserve the overall rank structure and variance allocation across conceptual dimensions.

We further examine dominant conceptual directions with cosine-similarity heatmaps between the top 100 eigenvectors of synthesized and natural co-occurrence matrices (Fig. 20). All models exhibit partial diagonal alignment, implying overlap in principal concept axes, while off-diagonal entries

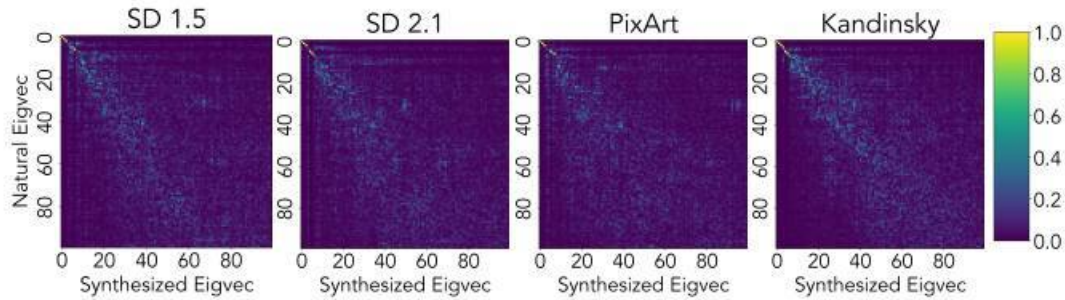


Figure 20: ● **Concept Basis Similarity.** Cosine similarity heatmaps between the top 100 eigenvectors of the natural and synthesized co-occurrence matrices ZZ^T . Diagonal structure indicates alignment of dominant conceptual directions, with varying degrees of alignment across the four models.

1009 reveal rotations and mismatches in higher modes. These patterns indicate structured deviations in
 1010 compositional geometry.

1011 Together, these findings show that diffusion models approximate the global shape of concept co-
 1012 activation, yet deviate in subtle and structured ways under spectral analysis. Such higher-order
 1013 discrepancies may contribute to failures in coherent multi-object scenes or relational concepts.

1014 G Additional Results: Blindspot Stress Testing

1015 To stress-test blindspots identified by the method, we collected prompts describing the target concepts
 1016 and used them to generate many images. We then compared outputs from models where the concept
 1017 was identified as a blindspot against outputs from models where it was not.

1018 Specifically, ChatGPT-4o was prompted as follows: *I want to generate an image of the following*
 1019 *concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input for a*
 1020 *text-to-image model. Return these as a list of strings in Python.* For each prompt, five images were
 1021 generated and analyzed with the custom RA-SAE model (Appendix B), which ranked them by the
 1022 intensity of the desired concept. All images were then manually reviewed to determine whether the
 1023 blindspot was successfully depicted.

1024 In the suppressed-concept examples below, partial aspects of the target sometimes appear, such as a
 1025 holder or string for bird feeder and a round hole for glossy DVD disc. However, the models
 1026 generally fail to generate the full concept, aligning with the method’s assessment and supporting the
 1027 stress-test result.

1028 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Images generated from prompts involving the bird feeder concept during stress testing. In Kandinsky, the method identifies this concept as a suppressed conceptual blindspot, matching the observed behavior that the model does not generate a corresponding image. By contrast, SD 1.5, where this concept was not identified as a blindspot, generates the concept.



Figure 22: Images generated from prompts involving the glossy DVD disc concept during stress testing. In SD 1.5, the method identifies this concept as a suppressed conceptual blindspot, matching the observed behavior that the model does not generate a corresponding image. By contrast, Kandinsky, where this concept was not identified as a blindspot, generates the concept.

1030 **H Additional Examples: Synthesized Images**



Figure 23: Additional LAION-5B image-caption examples with matching images generated from the same prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

1031 I Concentration bounds for δ

1032 In the experiments, $\delta(k)$ is estimated with $n = 10,000$ paired samples for each concept. Although
 1033 this budget is modest relative to the size of the underlying data distribution, it can still support reliable
 1034 concept-level estimates under boundedness assumptions. We formalize this with a concentration
 1035 bound on the empirical estimator $\hat{\delta}_n(k)$ using McDiarmid’s inequality [142], showing that concept-
 1036 bias estimates concentrate quickly with the evaluated sample size.

1037 **Theorem 4** (Concentration of $\hat{\delta}_n(k)$). *We assume that the concept score $\xi_k(\mathbf{x})$ takes values in $[a, b]$
 1038 almost surely for all images $\mathbf{x}$ drawn from either $\mathcal{D}_X$ or $\mathcal{D}'_X$. Let n paired samples $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$ be
 1039 drawn independently with $\mathbf{x}_i \sim \mathcal{D}_X$ and $\mathbf{x}'_i \sim \mathcal{D}'_X$, and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left(\frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1040 Let $M := b - a$ and $L := M/4$. Then for every $\varepsilon > 0$, the deviation satisfies

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right).$$

1041 *Proof.* The map $x \mapsto \sigma(x)$ is $1/4$ -Lipschitz because $|\sigma'(x)| \leq 1/4$ for all x .

1042 Treating $\hat{\delta}_n(k)$ as a function of the $2n$ independent variables $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$, changing
 1043 one argument changes the inner difference of means by at most M/n , and the outer sigmoid scales
 1044 this change by at most $1/4$. The bounded-difference constant for each coordinate is therefore
 1045 $(M/n)(1/4) = L/n$.

1046 By McDiarmid’s inequality [142],

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right),$$

1047 which proves the claim. □

1048 In practice, most concept scores $\delta(k)$ are sparse, with the majority concentrated near zero and only
 1049 a few reaching values up to 10. The concentration bound shows that, even for the largest observed
 1050 biases, $n = 10,000$ samples yield estimates of $\hat{\delta}_n(k)$ that deviate from the true value by at most
 1051 a small ε with high probability. This supports the sampling strategy used in the experiments and
 1052 indicates that concept-bias measurements are attainable with the evaluated sample size.

1053 J Monotonicity and Calibration-Free Interpretation of δ

1054 When analyzing blindspots, the practical goal is to rank concepts by the severity of their generative
 1055 bias. We use $\delta(k)$ for this ranking, but it is useful to verify that the score does not distort more
 1056 direct quantities such as the energy gap or odds ratio. The following result shows that $\delta(k)$ is a
 1057 strictly increasing reparameterization of both quantities, so it preserves their ordering and requires no
 1058 additional calibration for concept ranking.

1059 **Theorem 5** (Monotonicity and Calibration of $\delta_{g_\theta \leftrightarrow G}$). *For every concept index k define the energy*
 1060 *gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_X}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_X}[\xi_k(\mathbf{x})],$$

1061 *the associated odds ratio $\rho_k = \exp(\Delta_k)$, and the energy-difference score*

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

1062 *Then $\delta(k)$ is a strictly increasing bijection of both Δ_k and ρ_k , so ranking concepts by any one of*
 1063 *$\delta(k)$, Δ_k , or ρ_k produces exactly the same ordering.*

1064 *Proof.* The logistic sigmoid satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$; hence σ and therefore
 1065 $\delta(k) = \sigma(\Delta_k)$ grow strictly with Δ_k . Because the exponential map is also strictly increasing and
 1066 bijective $\mathbb{R} \rightarrow (0, \infty)$, setting $\rho_k = \exp(\Delta_k)$ preserves order and gives $\Delta_k = \log \rho_k$. Substituting
 1067 this identity into σ yields $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$, which is the composition of two
 1068 strictly increasing bijections and is therefore itself strictly increasing and bijective in ρ_k . Since strict
 1069 monotonic functions never reverse inequalities, the three quantities share the same total order over
 1070 concepts.

1071

□

1072 Thus, ranking concepts by $\delta(\cdot)$ is strictly equivalent to ranking them by energy gap or conceptual
 1073 generation odds ρ_k . No calibration is needed, because all three quantities preserve the same total
 1074 ordering over concepts.

K Stability of FID Under SAE Embeddings

In this section we establish a quantitative relationship between the Fréchet Inception Distance (FID) computed in the original activation space (of dimension d) and the FID after applying a (potentially overcomplete) SAE dictionary $D \in \mathbb{R}^{k \times d}$ with $k \gg d$. Throughout we assume that D has orthonormal columns but is not necessarily square, i.e.

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

We start by recalling a simple fact: if D is not overcomplete, orthogonal and $k = d$, then we have an isometry between A and Z , implying that the FID is perfectly preserved. However, this case is not realistic, we will then turn to the overcomplete case, and show we can bound FID by the extreme singular value of D . We will work with the Wasserstein-2 metric $\mathcal{W}_2$, noting that FID is just $\mathcal{W}_2^2$ specialised to Gaussians.

For a probability measure μ on $\mathbb{R}^d$ we write $D_\# \mu$ for its push-forward under D , i.e. $D_\# \mu(z) = \mu(D^{-1}z)$. Denote by $\sigma_{\min}$ and $\sigma_{\max}$ the minimal and maximal singular values of D , equivalently the square-roots of the extremal eigenvalues of $DD^\top$:

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

Empirically one usually finds $\sigma_{\min}, \sigma_{\max} \approx 1$, but the proof does not rely on that. We will start by a simple lemma in the case where D is not overcomplete.

Lemma 1 (Isometry under exact orthogonality). *Suppose $k = d$ and $D^\top D = DD^\top = I_d$. Then D is an isometry: $\|Dv\|_2 = \|v\|_2$ for all $v \in \mathbb{R}^d$. Consequently, for any probability measures μ, ν on $\mathbb{R}^d$ with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

Proof. Orthogonality of D implies preservation of the Euclidean norm, and push-forward commutes with the map inside the $\mathcal{W}_2$ infimum; the integrand is unchanged, so the infimum value is identical. $\square$

This case, however, is quite unrealistic as SAE usually rely on the overcompleteness to extract meaningful and interpretable concepts. In the overcomplete case, D is no longer orthonormal, but we can still have column-orthonormal dictionary. We will use that to show that we can bound using the extremal singular value of $D^\top D$.

Theorem 6 (FID under column orthogonal embeddings). *Let $D \in \mathbb{R}^{k \times d}$ satisfy $D^\top D = I_d$ and denote by $0 < \sigma_{\min} \leq \sigma_{\max}$ the extreme singular values of $DD^\top$. Given two data matrices $A, A' \in \mathbb{R}^{n \times d}$ (rows are sample vectors), set $Z = AD^\top \in \mathbb{R}^{n \times k}$ and $Z' = A'D^\top \in \mathbb{R}^{n \times k}$. Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

Proof. Let μ denote the empirical measure of A and let ν denote the empirical measure of A' , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

For any coupling $\pi \in \Pi(\mu, \nu)$ (i.e. a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ, ν) we have, by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

Integrating with respect to an arbitrary coupling $\pi \in \Pi(\mu, \nu)$ yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

The middle integral is the transport cost of the pushed-forward coupling $(D \times D)_\# \pi$ between $\mu_D := D_\# \mu$ and $\nu_D := D_\# \nu$. Because the inequalities hold for every π , they also hold for

1109 couplings attaining $\mathcal{W}_2(\boldsymbol{\mu}, \boldsymbol{\nu})$ and $\mathcal{W}_2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D)$, even though these optima need not coincide. Taking
 1110 the infimum over $\boldsymbol{\pi}$ term-wise makes the comparison explicit:

$$\sigma_{\min}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi} \leq \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|D(\mathbf{x} - \mathbf{y})\|_2^2 d\boldsymbol{\pi} \leq \sigma_{\max}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi}.$$

1111 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}) \leq \mathcal{W}_2^2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

1112 Recognizing $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$ for the Gaussian surrogate, and substituting $(\mathbf{A}, \mathbf{A}')$ and $(\mathbf{Z}, \mathbf{Z}')$
 1113 respectively, finishes the proof. $\square$

1114 Theorem 6 shows that applying a column-orthogonal overcomplete SAE dictionary cannot distort
 1115 Fréchet Inception Distance by more than the square of its extremal singular values. When $\mathbf{D}$ is nearly
 1116 orthogonal—as we empirically found with $\sigma_{\min}, \sigma_{\max} \approx 1$ —FID measured in the SAE feature space
 1117 remains close to canonical FID. $\square$

1118